

A Portable Experimental Buoy for Coastal Acoustic Monitoring as a Potential Complement to the IMS Hydroacoustic Network: Applications on the Argentinian Continental Shelf

Prario, I; Bos, P; Marques Rojo, R; Cinquini, M; Blanc, S.

Argentinian Navy Research Office & UNIDEF (CONICET)

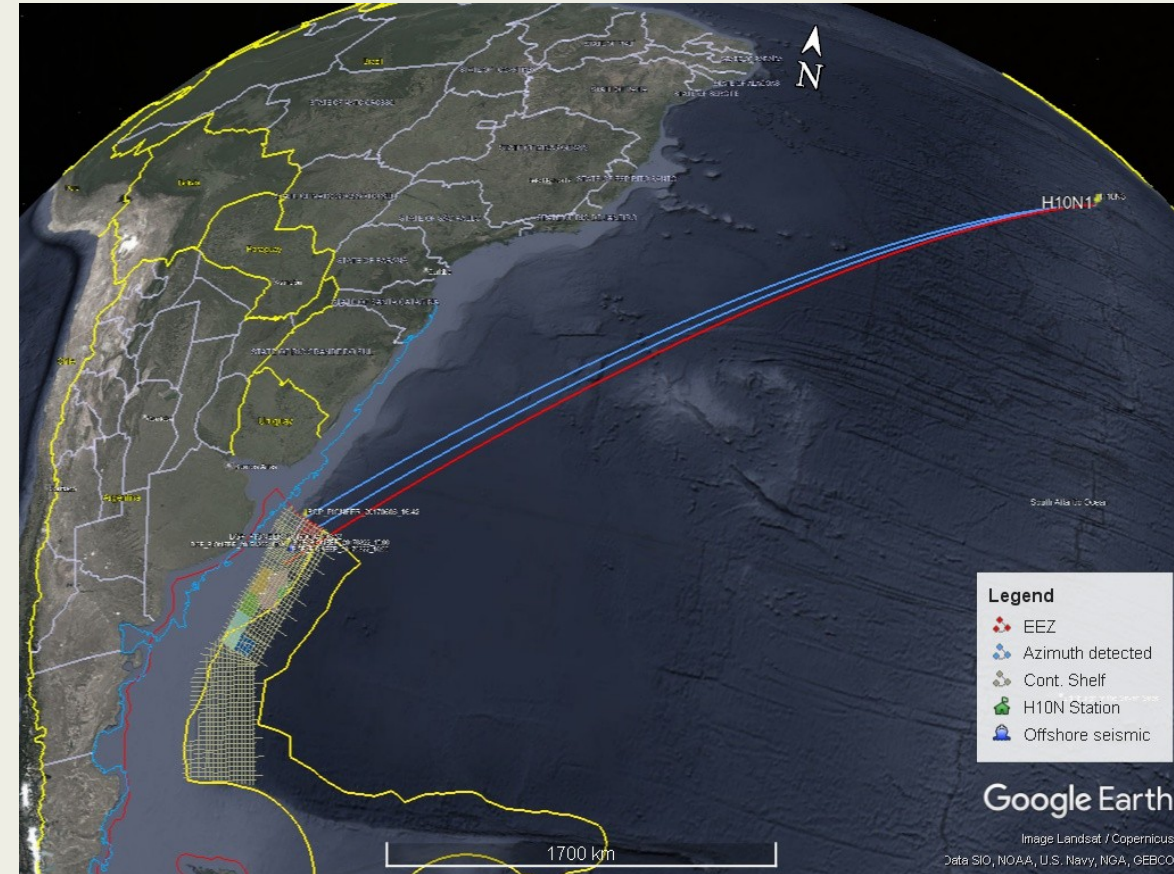


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Presentation Date: 11 September 2025

Background

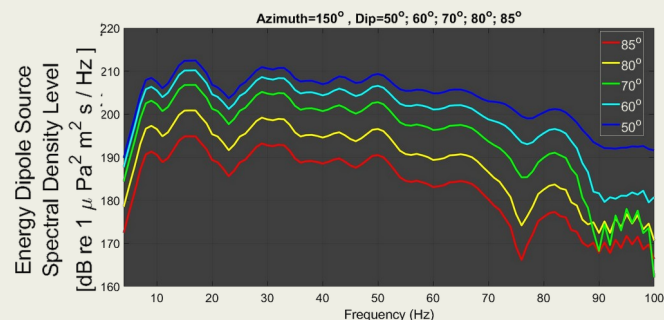
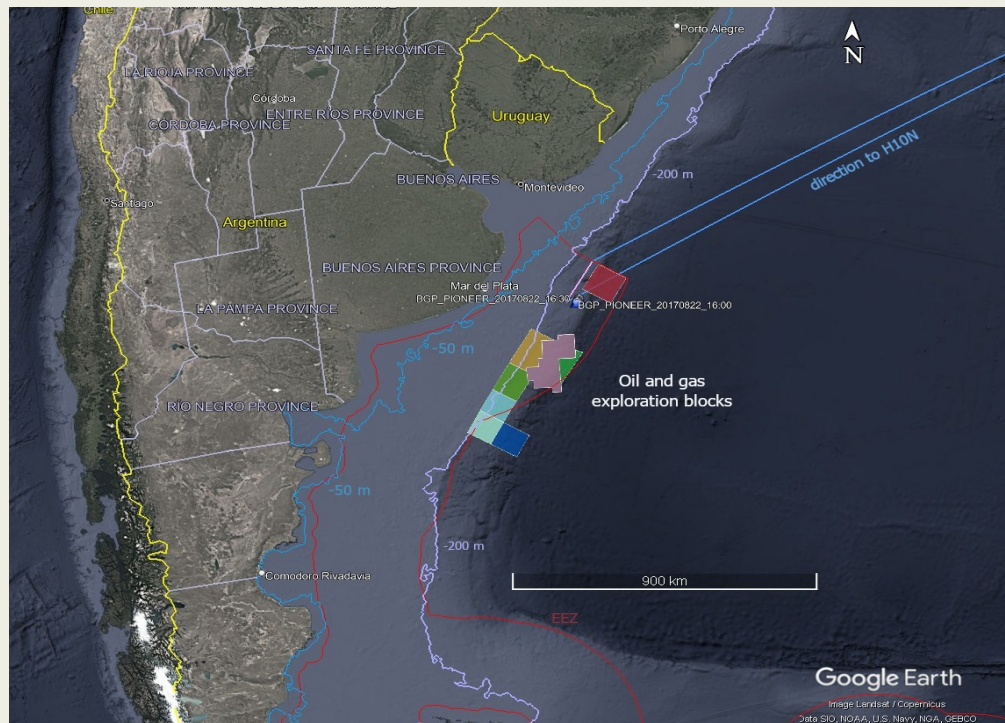
- Airgun array used in seismic surveys generate sound pulses 5 Hz –60 Hz that can propagate over ocean-basin scales.
- Under favourable conditions, signals reach CTBT IMS hydroacoustic (HA) stations with high SNR.
- On the Argentinian Continental Shelf, airgun signals detected at H10N can be used to validate long-range propagation models ⁽¹⁾.
- Limitations: highly directional sources and shallow-water propagation reduce detectability.
- Challenge: synchronize ground-truth airgun shots with H10N recordings, actual Source Levels.



⁽¹⁾ SnT2021 P1.3-494 An inverse problem approach for acoustic Transmission Loss estimation from the analysis of signals generated by seismic air-gun arrays.
SnT2023 O1.3-536 Analysis of bathymetric effects on long-range acoustic propagation using 2D and 3D Transmission Loss models in signals generated by airgun arrays.

Motivation

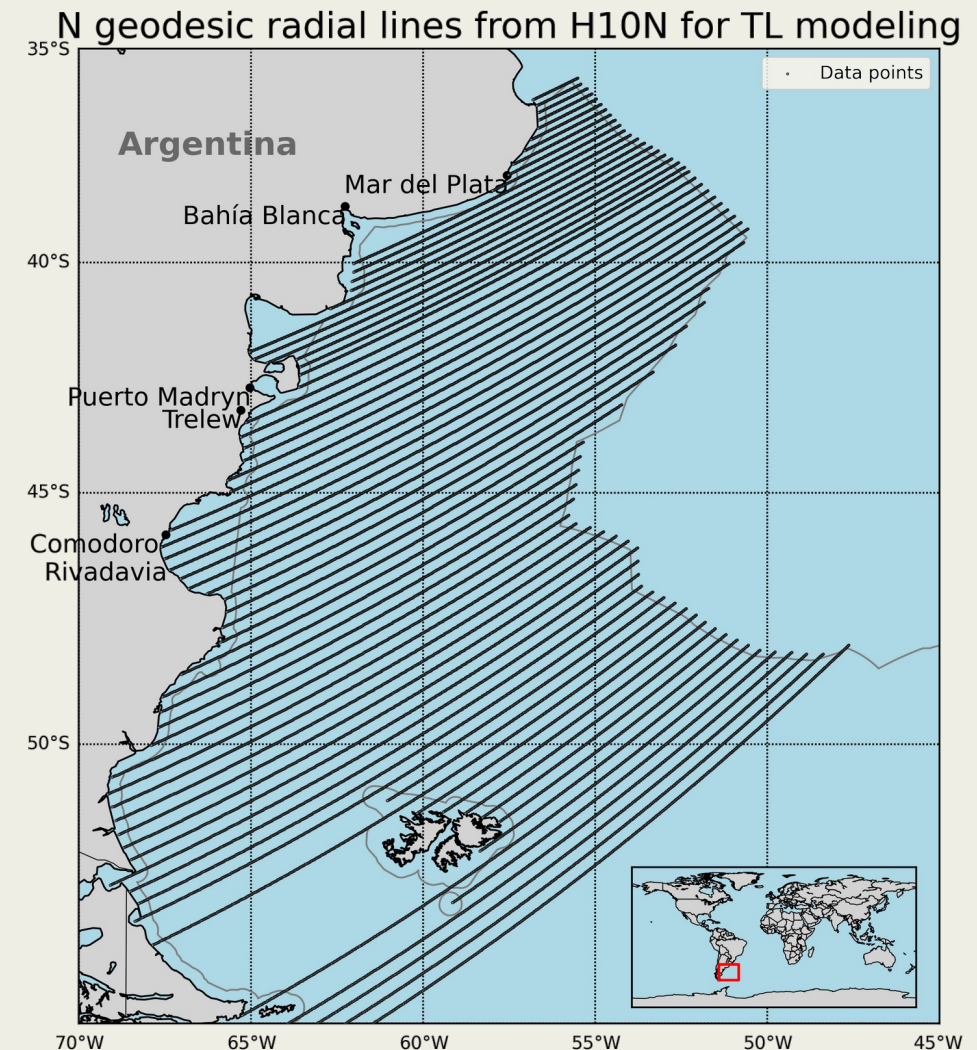
- Seismic surveys are conducted in offshore oil & gas exploration blocks.
- Large dataset of high-amplitude impulsive signals
▶ basin-scale propagation analysis.
- Airgun array source levels simulated with an open-source airgun modelling software⁽²⁾: Energy source level $\sim 190 - 210$ dB re $1 \mu\text{Pa}^2\text{m}^2\text{s}$ within 5 Hz – 70 Hz band (HA10 direction).
- Need for local measurement of airgun shots signals or other underwater noise sources on the Continental Shelf ▶ **selection of semi-permanent sites for buoy-deployed hydrophones.**



(2) Sertlek, H. O., and Ainslie, M. A. (2015). "AGORA: Airgun Source Signature Model: Its application for the Dutch Seismic Survey," in Proceedings of Underwater Acoustics Conference and Exhibition, Crete, Greece, pp. 439–446,

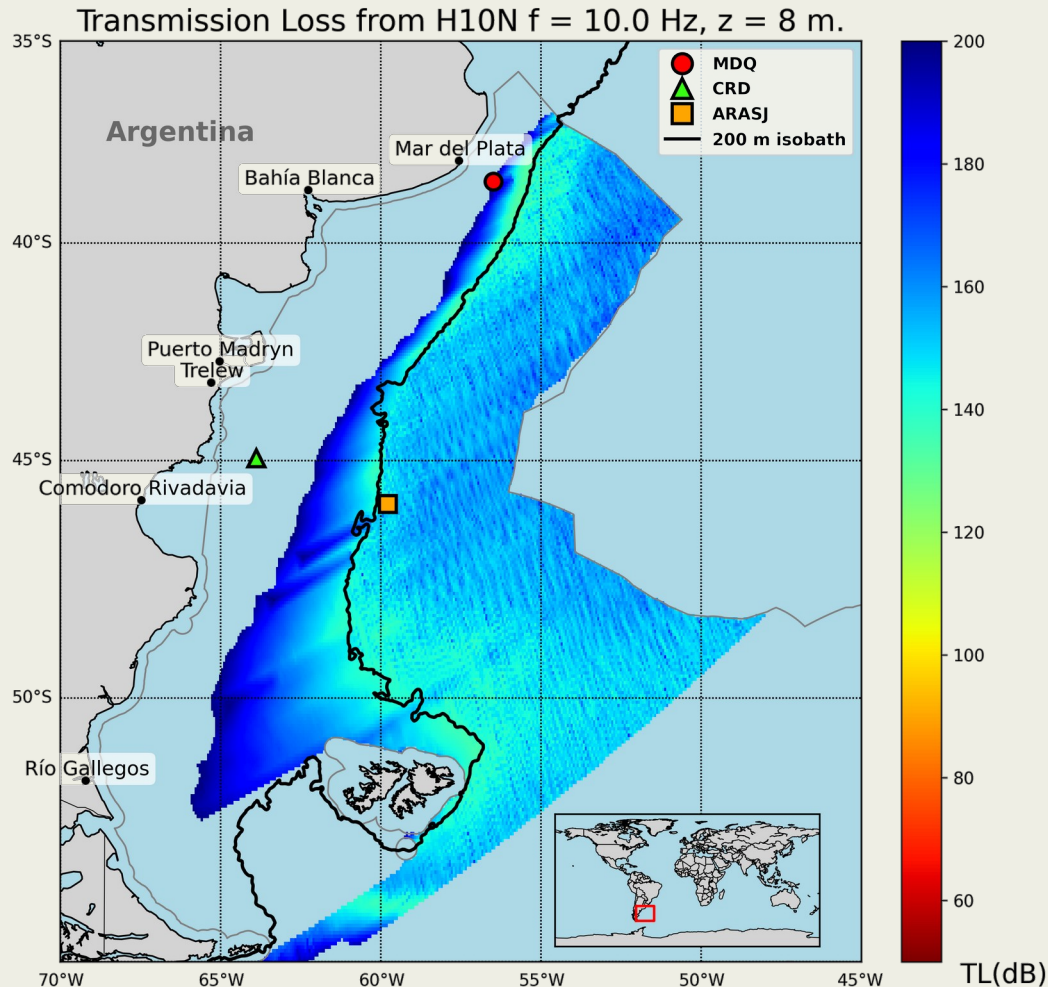
Current acoustic coverage in Argentinian Continental Shelf

- **Goal:** identify candidate shallow-water sites (depth $z < 200$ m) for improved local monitoring.
- **Acoustic coverage areas of the HA10 station:** preliminary analysis using 2D Parabolic Equation (PE) model RAM (Range-dependent Acoustic Model, Julia implementation).
- **Acoustic scenario:**
 - Oceanographic databases: T, S profiles from Copernicus; bathymetry from GEBCO.
 - Frequencies band analyzed: $2 \text{ Hz} < f < 100 \text{ Hz}$.
 - Transmission Loss (TL) computed along Nx2D radial transects from H10N.
 - Source depth = 8 m (typical for airgun array).
- **Maximum coverage analysis:** High impulsive source levels⁽³⁾; $TL < 200 \text{ dB}$; $NL \sim 80 \text{ dB re } 1 \mu\text{Pa}^2$.



⁽³⁾ Dekeling, R.P.A., et al. Monitoring Guidance for Underwater Noise in European Seas, Part III

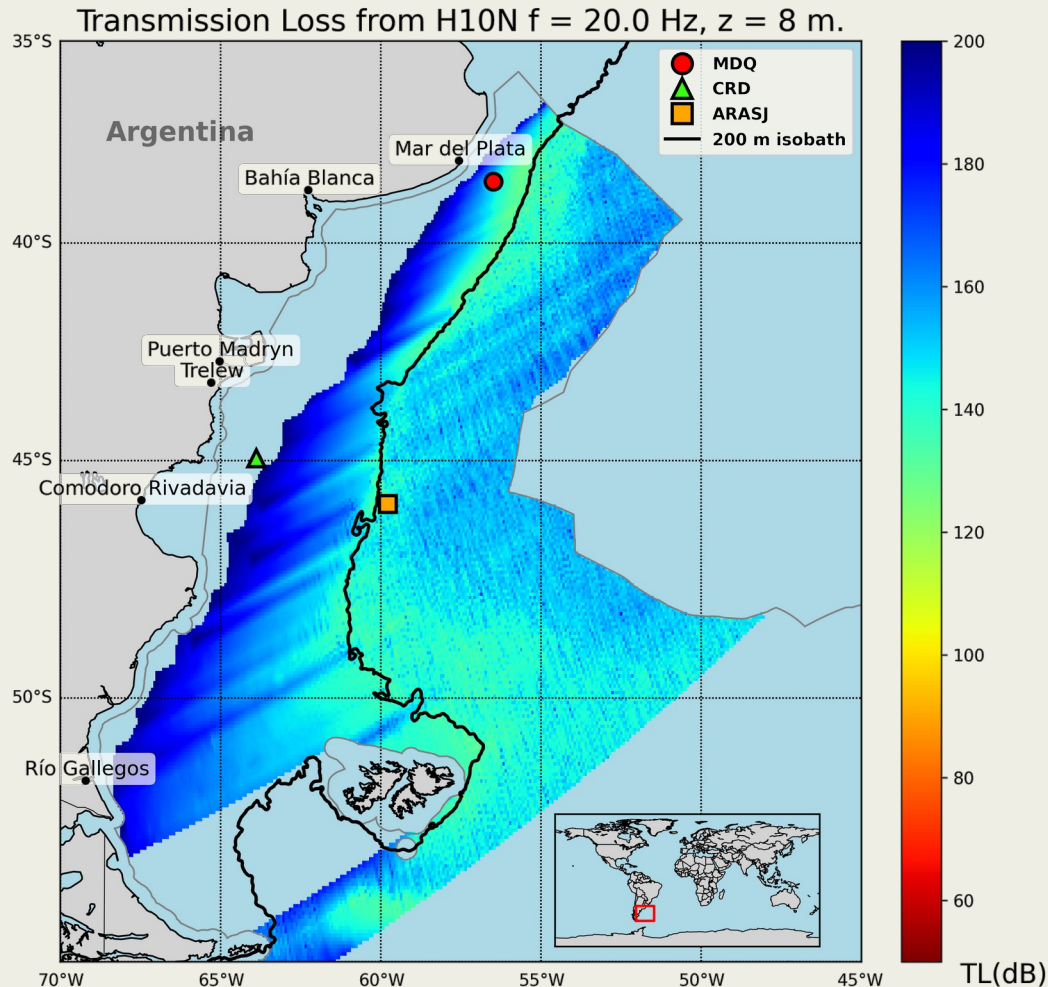
Offshore Argentina coverage of Ascension station H10N



- Strong frequency dependence:
 - Low frequencies ($f < 10$ Hz): coastal shadow zones, $TL > 200$ dB.
 - Higher frequencies ($10 \text{ Hz} < f < 80 \text{ Hz}$): partial coverage, but limited in shallow waters.
- Argentinian continental slope acts as an acoustic filter.
- Coastal coverage gaps: long-range IMS hydroacoustic stations miss nearshore regions.
- Local minima of TL in the continental slope (ARA San Juan wreckage, signal of 2017 implosion event was detected in two IMS-HA stations⁴)
- Offshore candidate sites identified: 1-MDQ, 2-CRD

⁽⁴⁾ Prario, I., Cinquini, M., Marques Rojo, R. *et al.* Characterization of the Acoustic Event Associated with the Loss of the ARA San Juan Submarine Based on Long-Range Measurements by CTBTO's Hydrophone Stations. *Pure Appl. Geophys.* **180**, 1317–1342 (2023). <https://doi.org/10.1007/s00024-022-03090-0>.

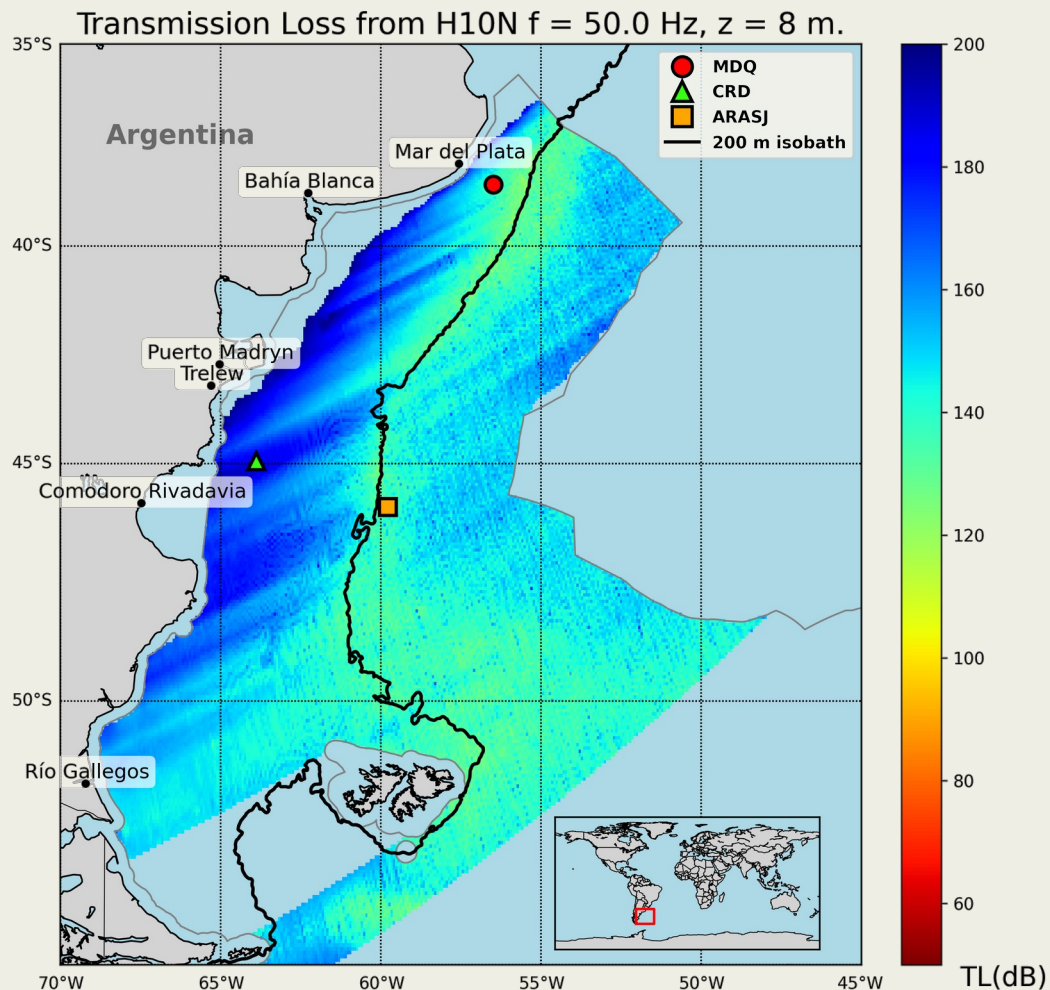
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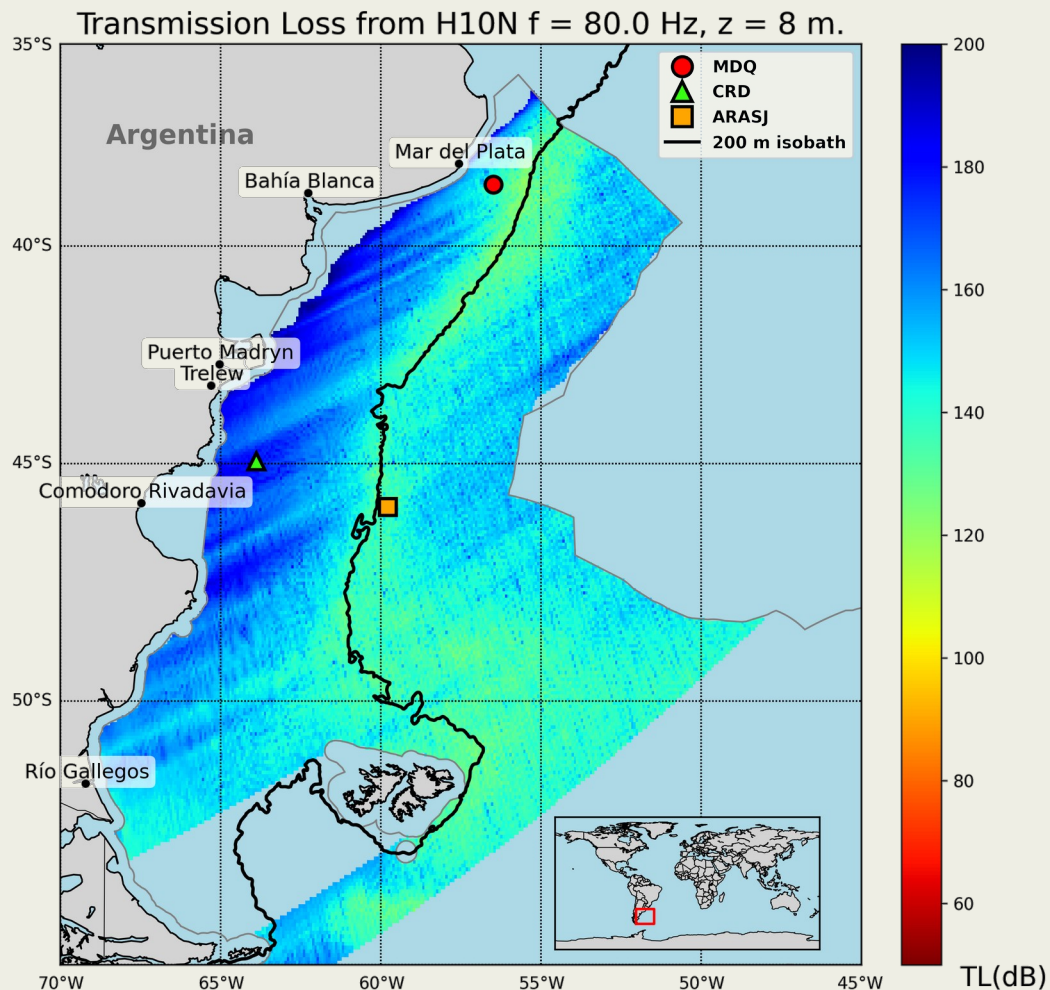
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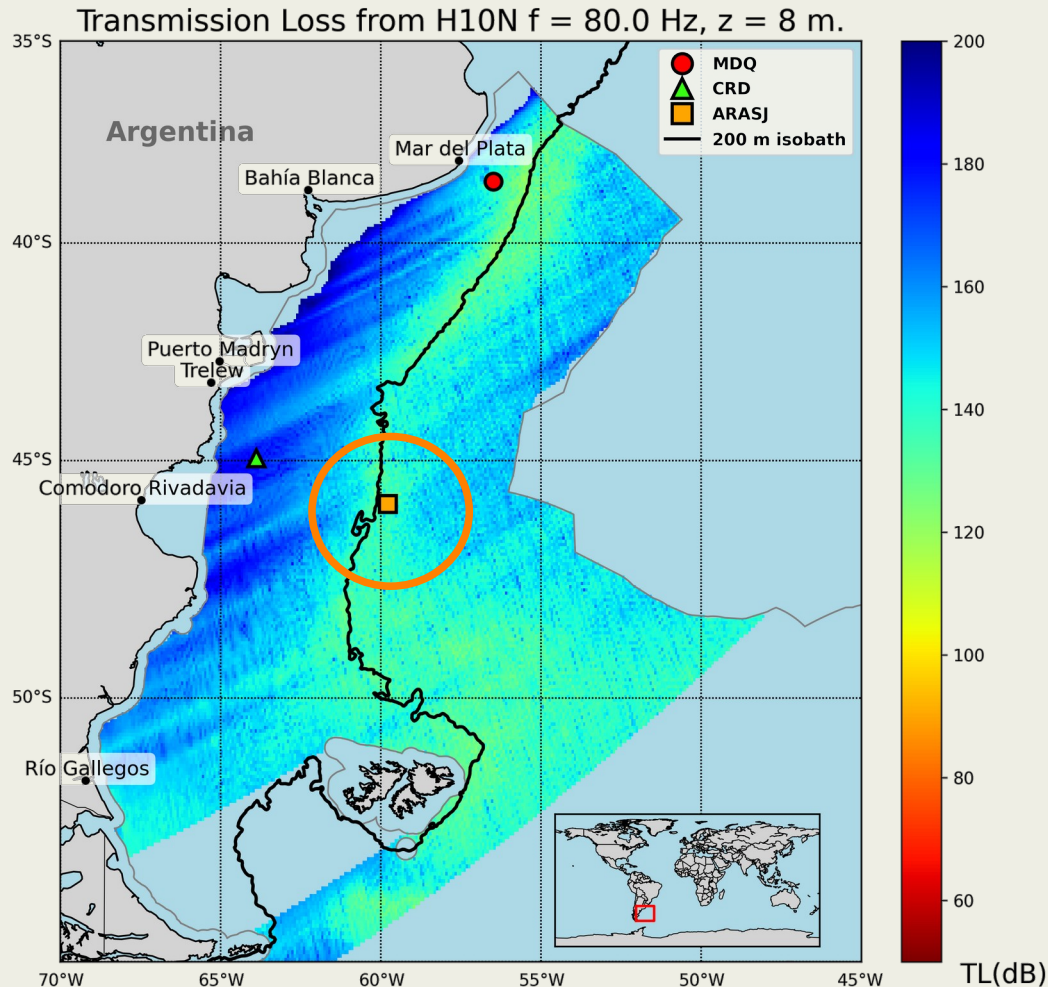
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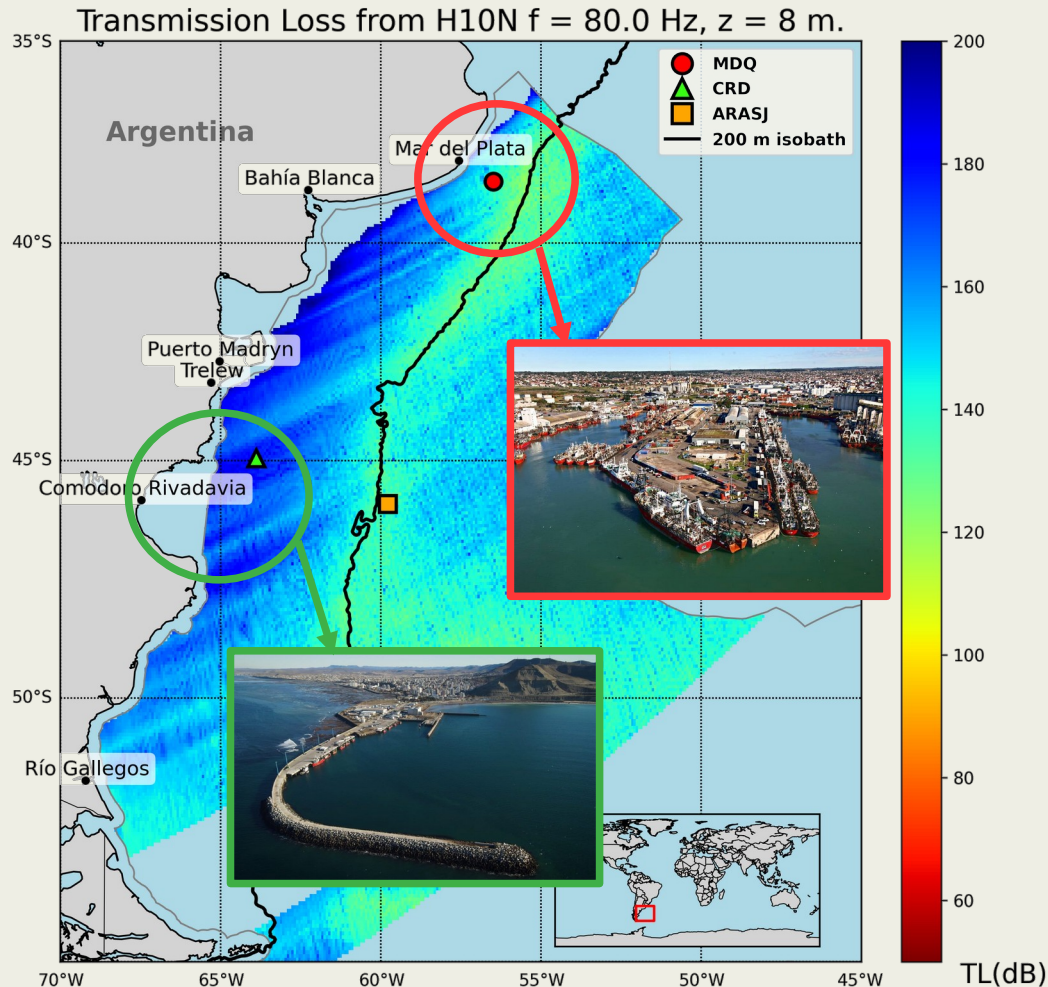
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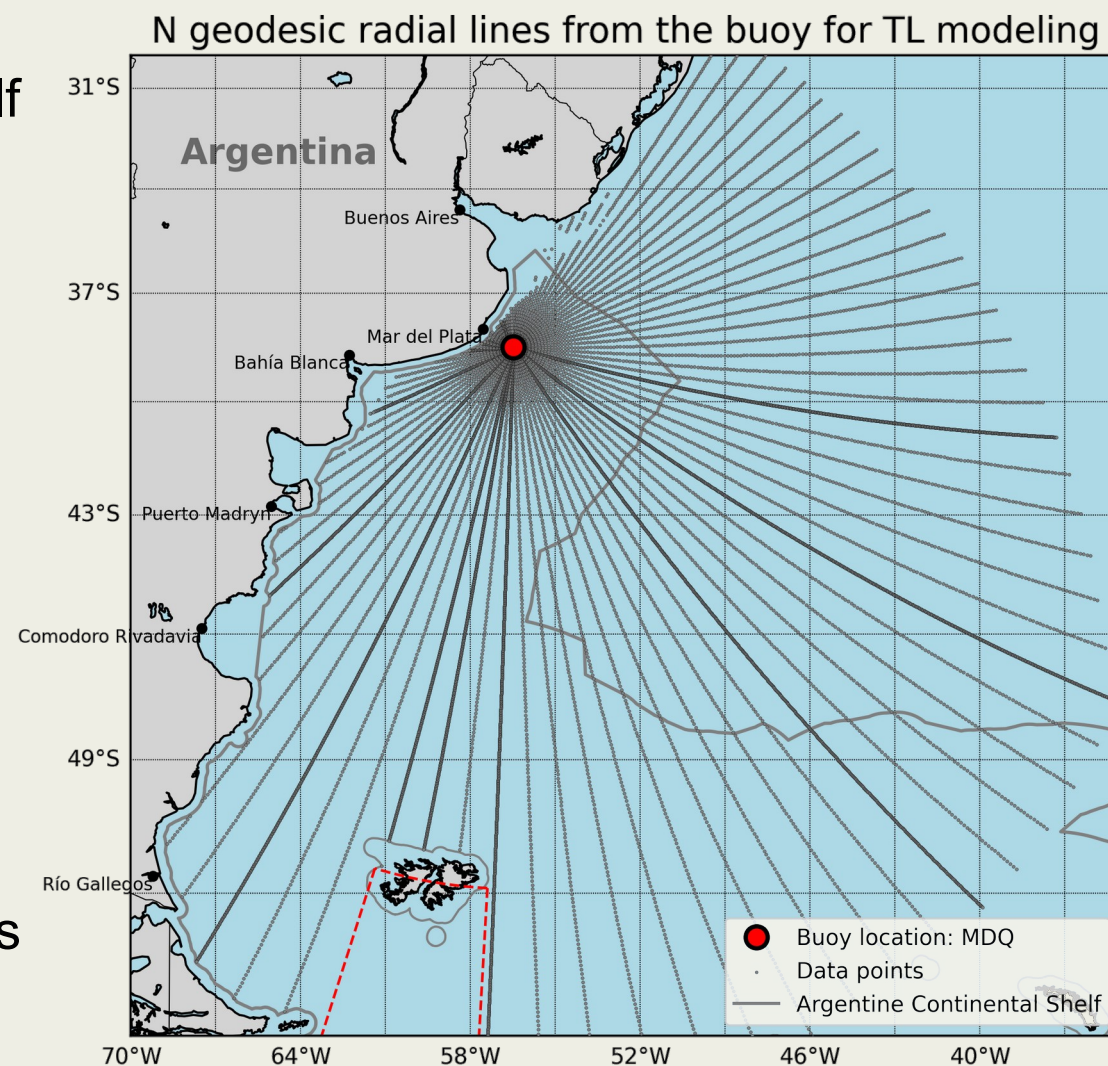


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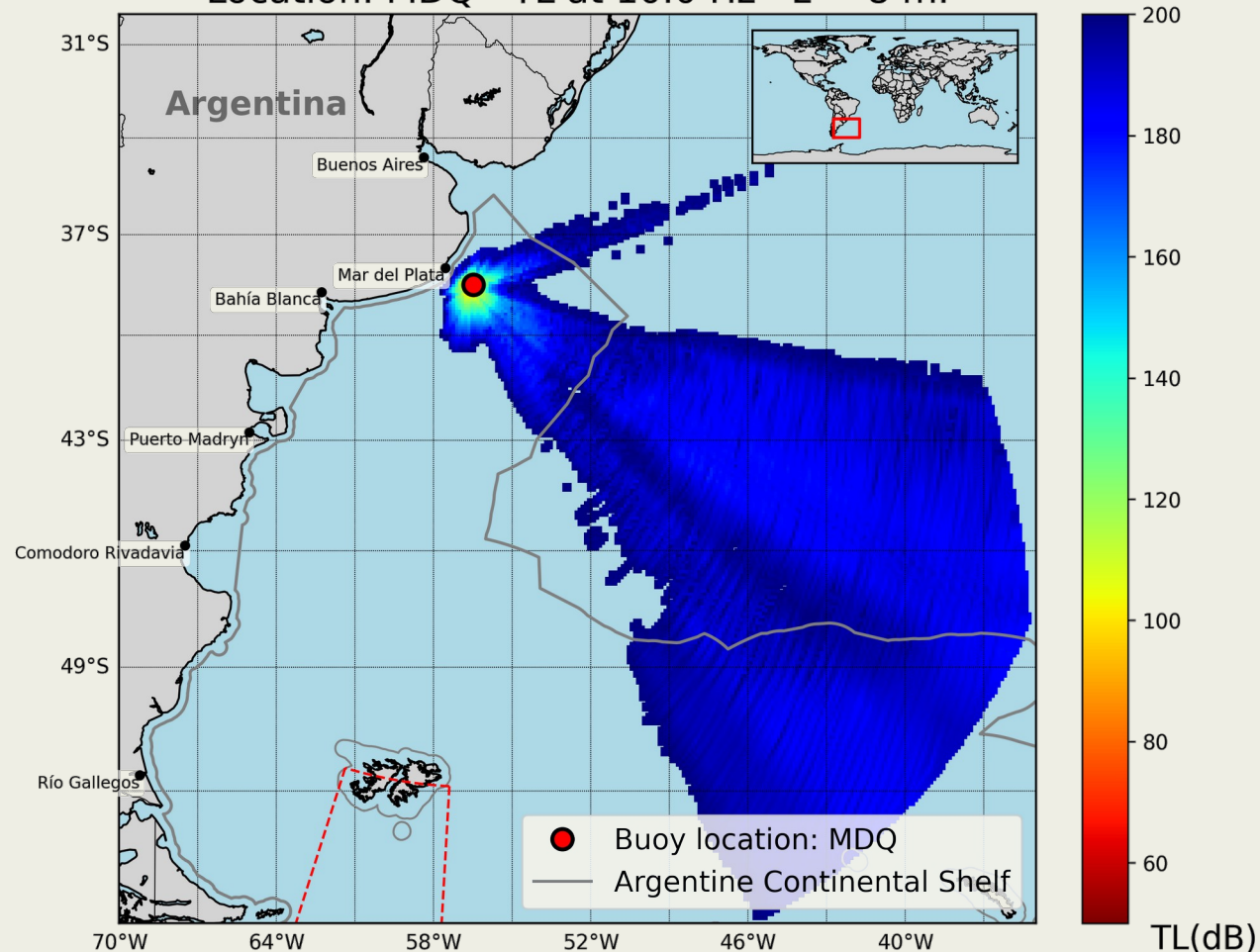
Coverage of **candidate 1** station (offshore Mar del Plata - **MDQ**)

- **Specific goal:**
 - Assess acoustic coverage over the continental shelf and adjacent waters.
- **Acoustic scenario:**
 - Frequency range: 2 Hz – 100 Hz (results shown at 10, 20, 50, 80 Hz).
 - Source depth = 8 m (typical airgun tow depth).
 - Receiver depth (hydrophone) = 40 m.
 - T, S profiles from Copernicus.
 - Bathymetry from GEBCO database.
 - Seabed properties from *Jensen et al.*
 - NL~80-85 dB re $1\mu\text{Pa}^2$
- **Simulation:**
 - 2D PE with RAM.jl.
 - Transmission Loss (TL) along Nx2D radial transects from the candidate location.



Coverage of **candidate 1** station (offshore Mar del Plata - **MDQ**)

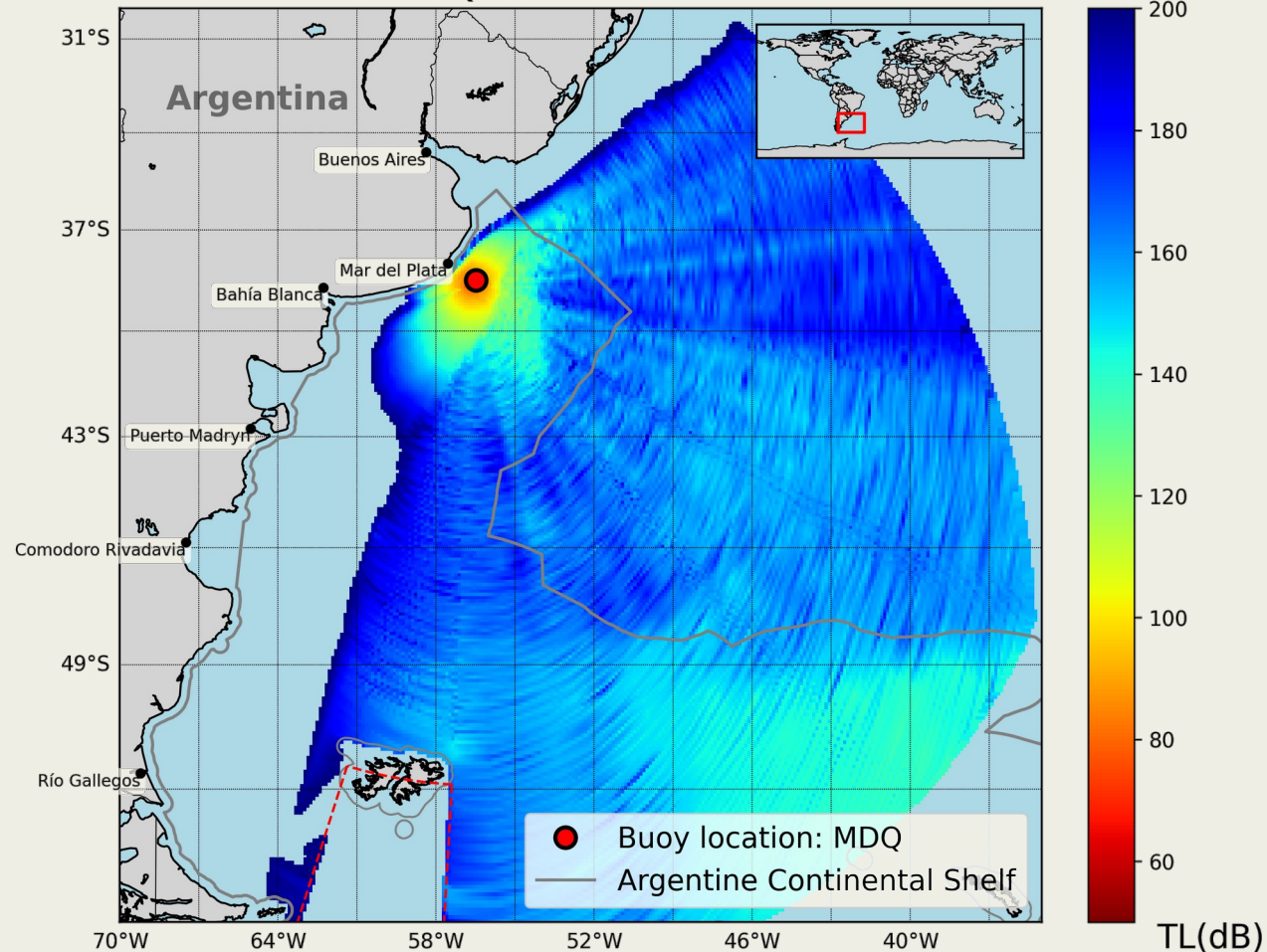
Location: MDQ - TL at 10.0 Hz - $z = 8$ m.



- Strong frequency dependence: low frequencies ($f < 10$ Hz) show larger shadow zones.
- Filtering effect: continental slope reduces transmission at low frequencies ($f < 20$ Hz).
- Improved coverage at $50 \text{ Hz} < f < 80 \text{ Hz}$ band across shelf and slope.
- Shallow-water gap: limited detectability close to the coast (depth $z < 200$ m).
- Implication: buoy-deployed hydrophones at MDQ station enhances monitoring in northern shelf region.

Coverage of **candidate 1** station (offshore Mar del Plata - **MDQ**)

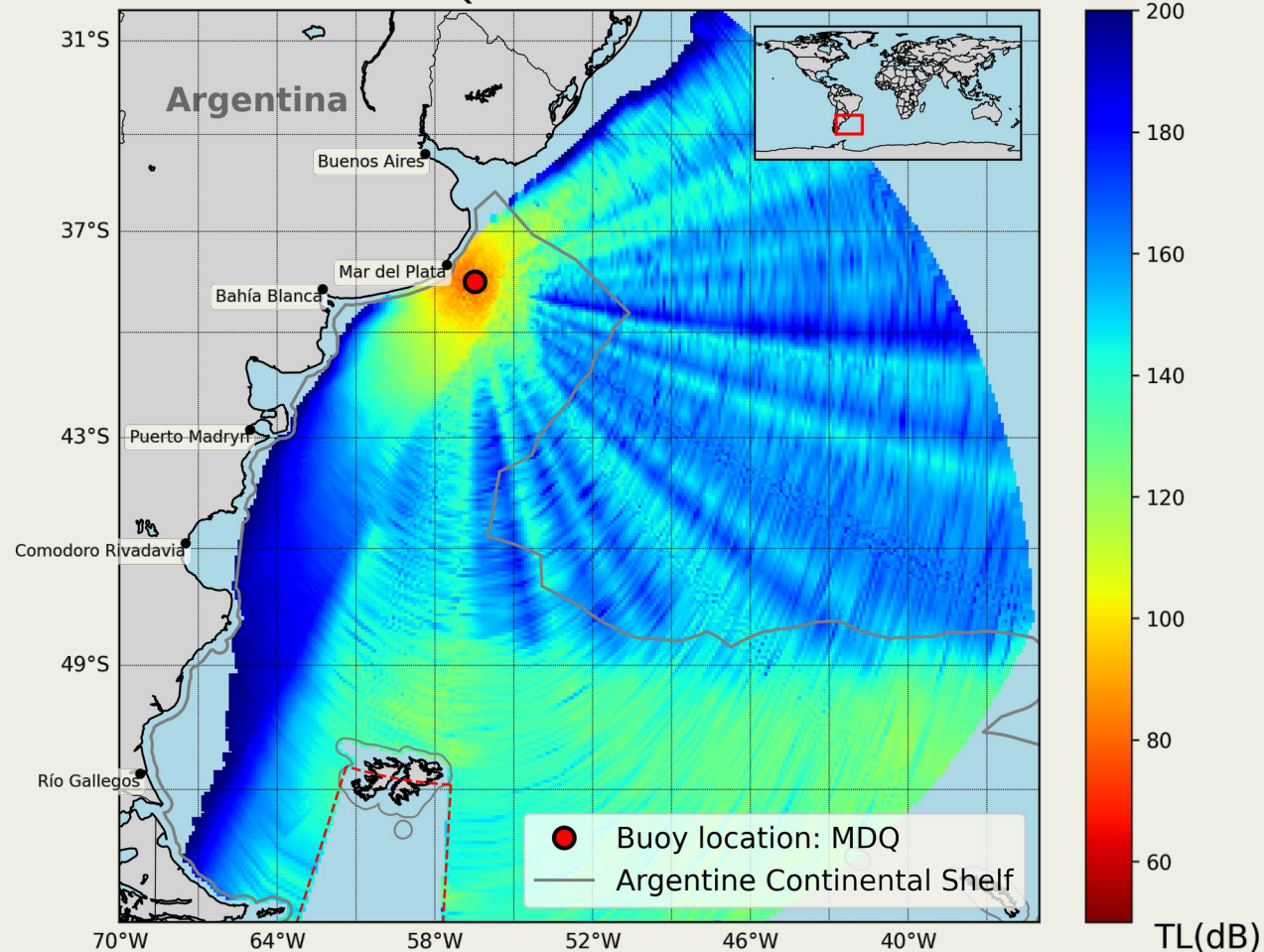
Location: MDQ - TL at 20.0 Hz - $z = 8$ m.



- Strong frequency dependence: low frequencies ($f < 10$ Hz) show larger shadow zones.
- Filtering effect: continental slope reduces transmission at low frequencies ($f < 20$ Hz).
- Improved coverage at $50 \text{ Hz} < f < 80 \text{ Hz}$ band across shelf and slope.
- Shallow-water gap: limited detectability close to the coast (depth $z < 200$ m).
- Implication: buoy-deployed hydrophones at MDQ station enhances monitoring in northern shelf region.

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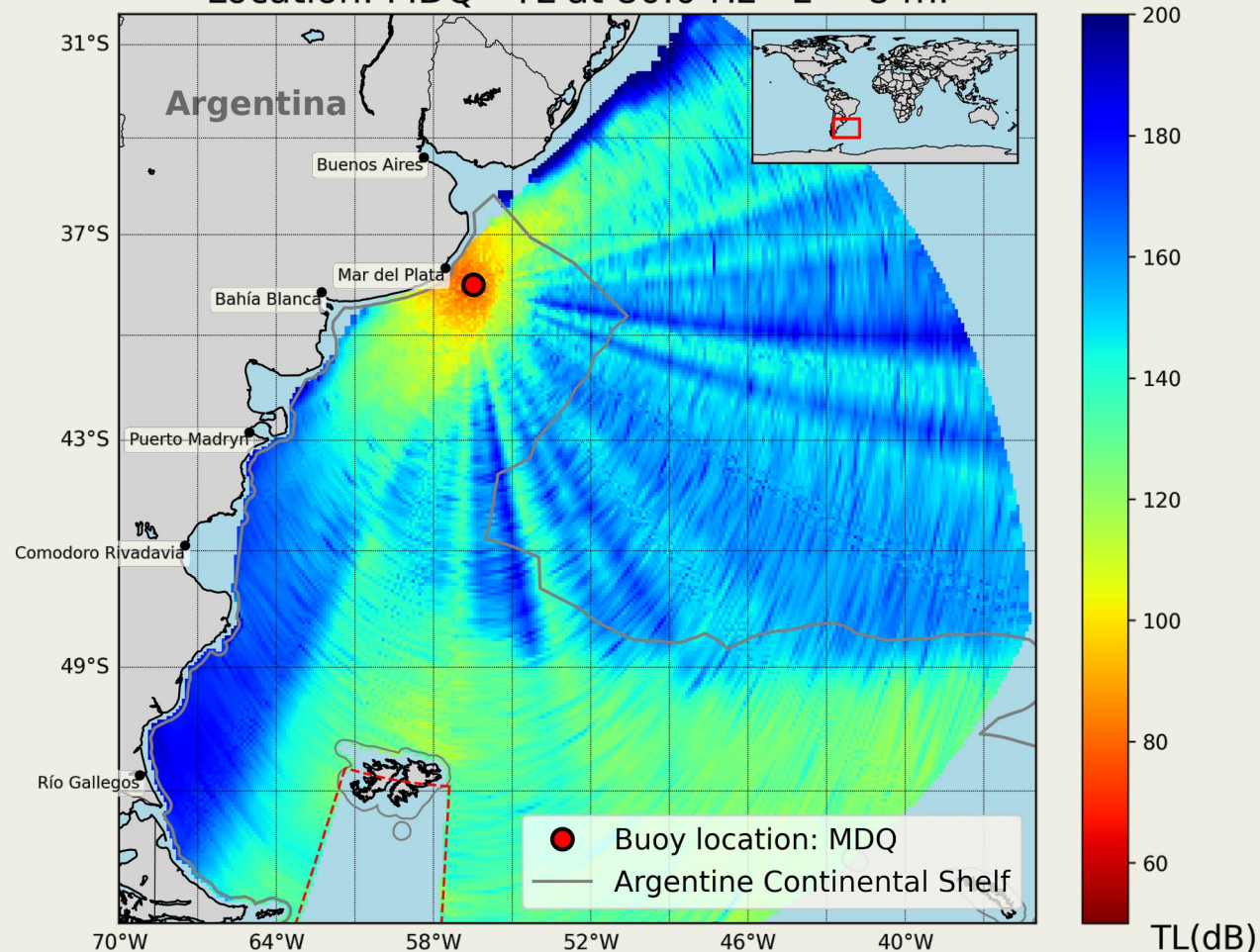
Location: MDQ - TL at 50.0 Hz - $z = 8$ m.



- Strong frequency dependence: low frequencies ($f < 10$ Hz) show larger shadow zones.
- Filtering effect: continental slope reduces transmission at low frequencies ($f < 20$ Hz).
- Improved coverage at $50 \text{ Hz} < f < 80 \text{ Hz}$ band across shelf and slope.
- Shallow-water gap: limited detectability close to the coast (depth $z < 200$ m).
- Implication: buoy-deployed hydrophones at MDQ station enhances monitoring in northern shelf region.

Coverage of **candidate 1** station (offshore Mar del Plata - **MDQ**)

Location: MDQ - TL at 80.0 Hz - $z = 8$ m.



- Strong frequency dependence: low frequencies ($f < 10$ Hz) show larger shadow zones.
- Filtering effect: continental slope reduces transmission at low frequencies ($f < 20$ Hz).
- Improved coverage at $50 \text{ Hz} < f < 80 \text{ Hz}$ band across shelf and slope.
- Shallow-water gap: limited detectability close to the coast (depth $z < 200$ m).
- Implication: buoy-deployed hydrophones at MDQ station enhances monitoring in northern shelf region.

Coverage of **candidate 2** station (offshore Comodoro Rivadavia - **CRD**)

Specific goal:

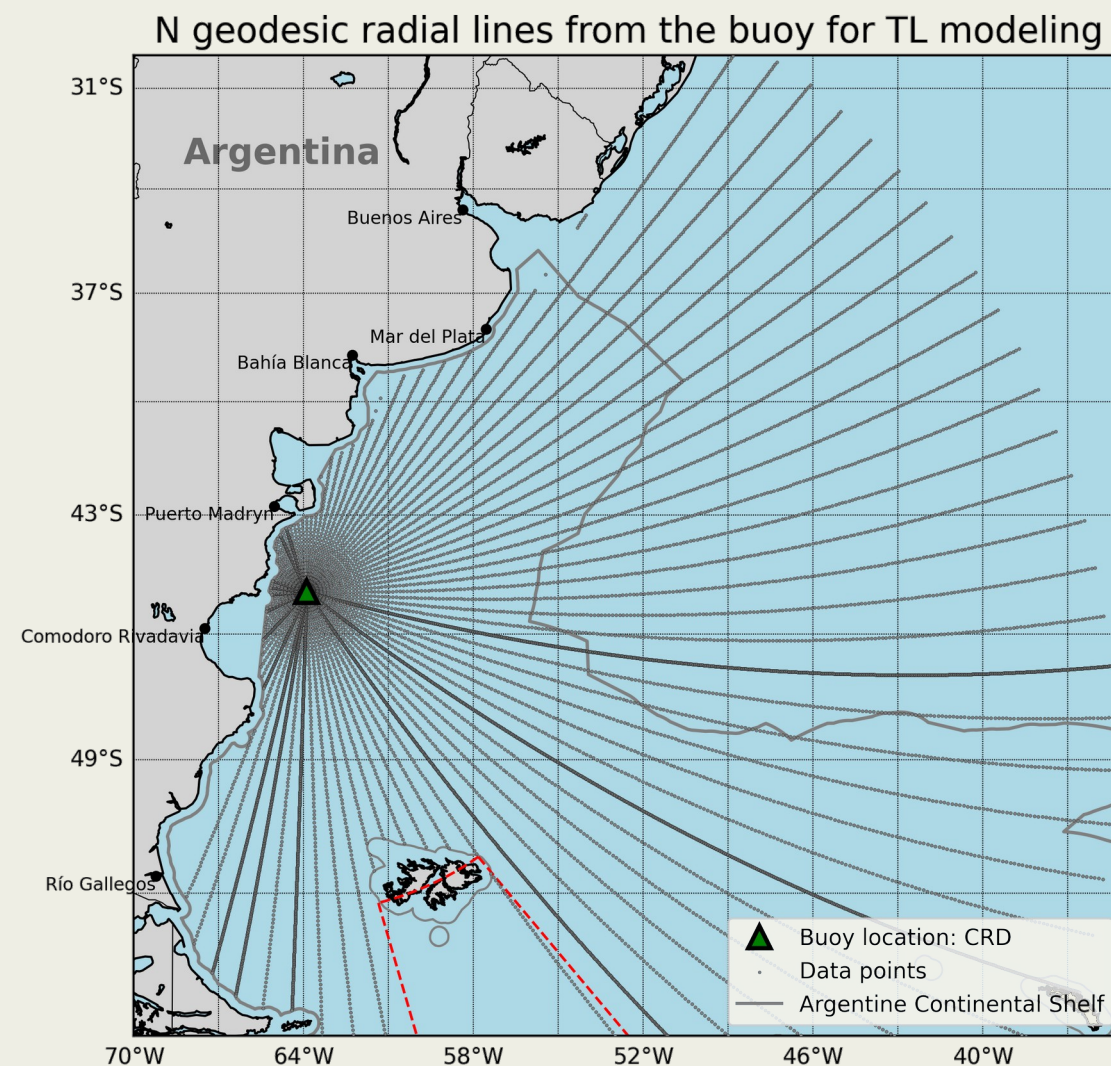
- Assess acoustic coverage over the continental shelf and adjacent waters.

Acoustic scenario:

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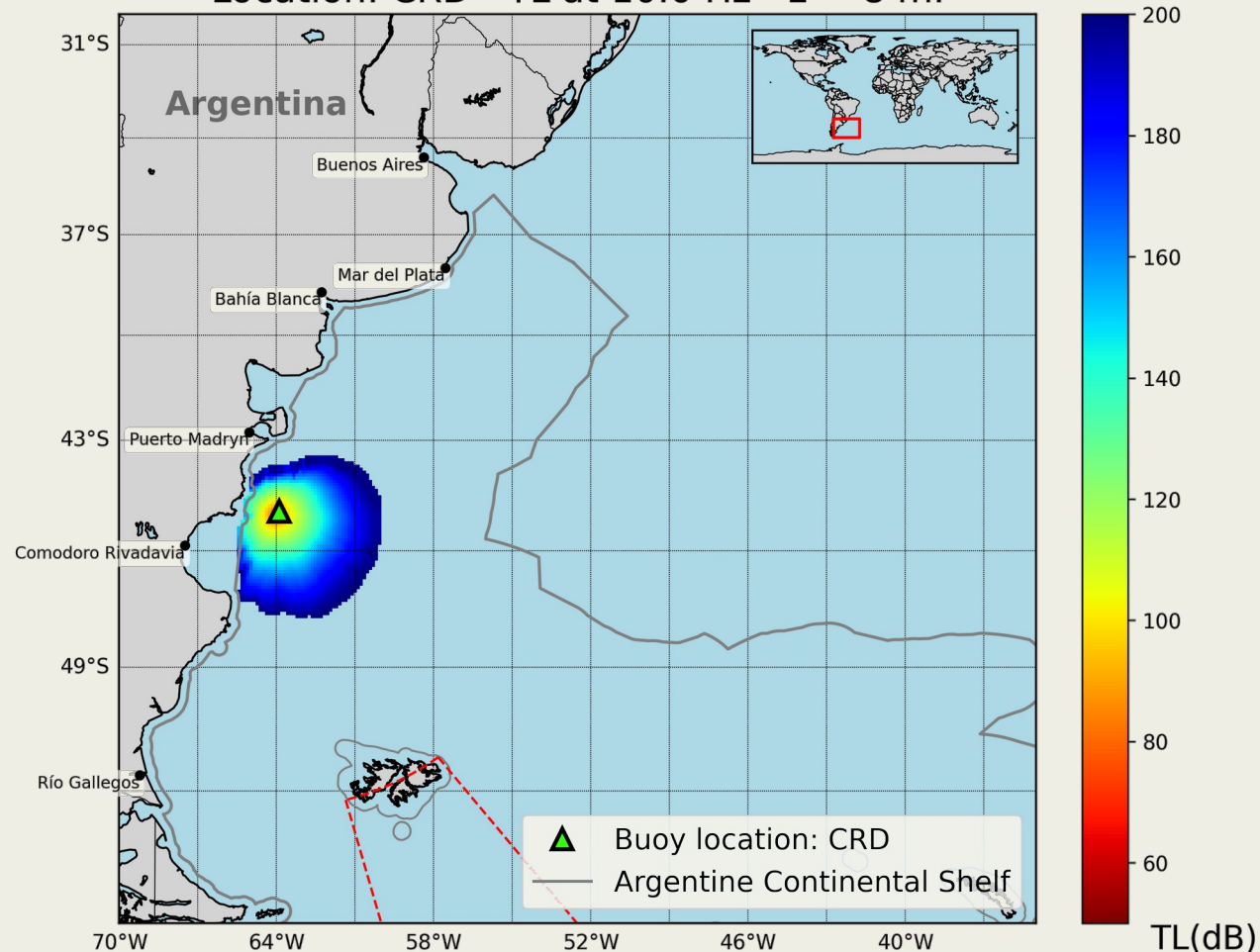
Simulation:

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- Transmission Loss (TL) along Nx2D radial transects from the candidate location.



Coverage of candidate 2 station (offshore Comodoro Rivadavia - CRD)

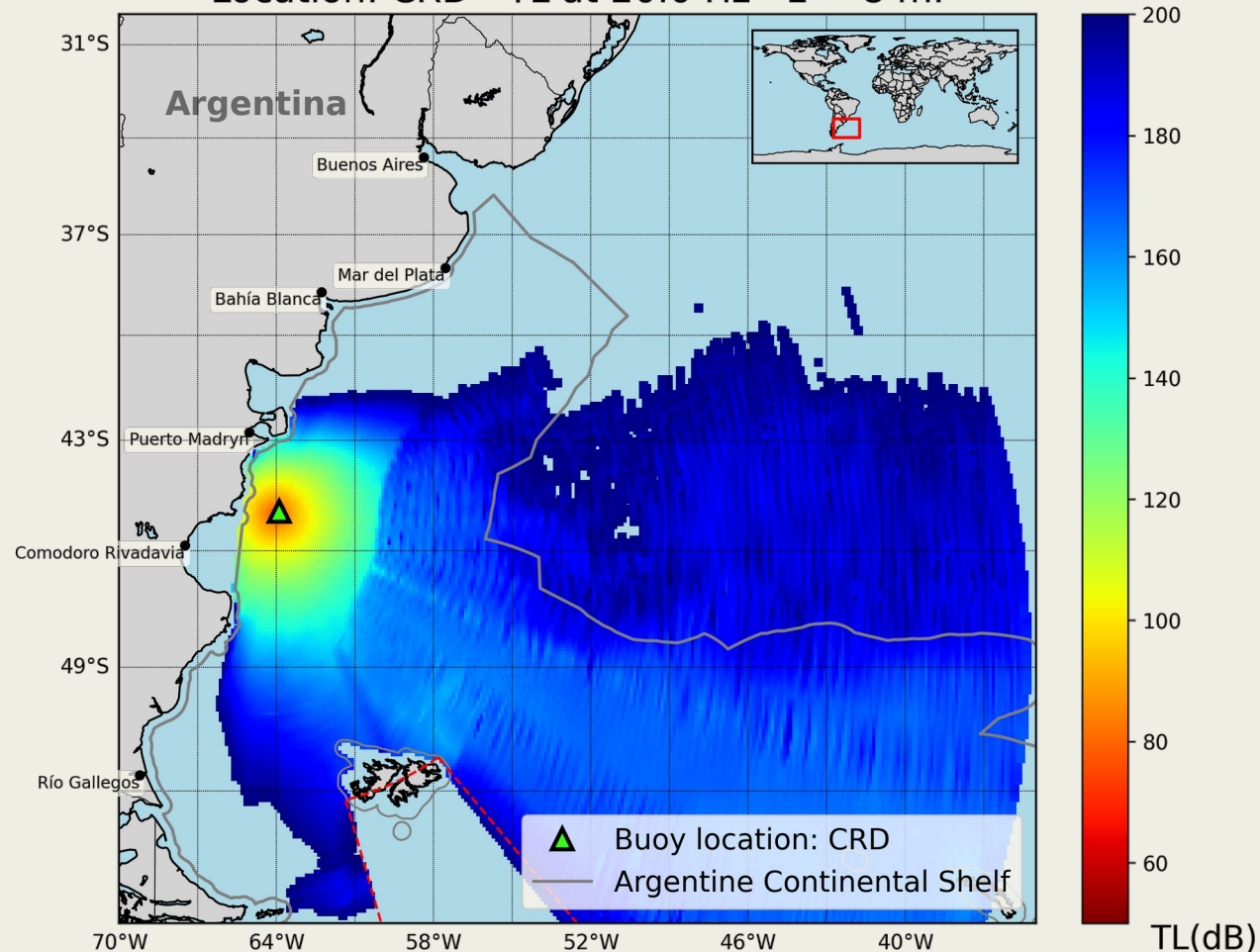
Location: CRD - TL at 10.0 Hz - $z = 8$ m.



- Marked frequency variation: stronger propagation loss at 10 Hz – 20 Hz band rather than at 50 Hz – 80 Hz band.
- Continental slope creates coastal shadow zones at low frequencies.
- Broader mid-frequency coverage compared to MDQ station, extending farther offshore.
- Shallow-water limitations remain near coastlines.
- Implication: buoy-deployed hydrophones at CRD station can improve coverage in southern shelf and can complement MDQ station.

Coverage of candidate 2 station (offshore Comodoro Rivadavia - CRD)

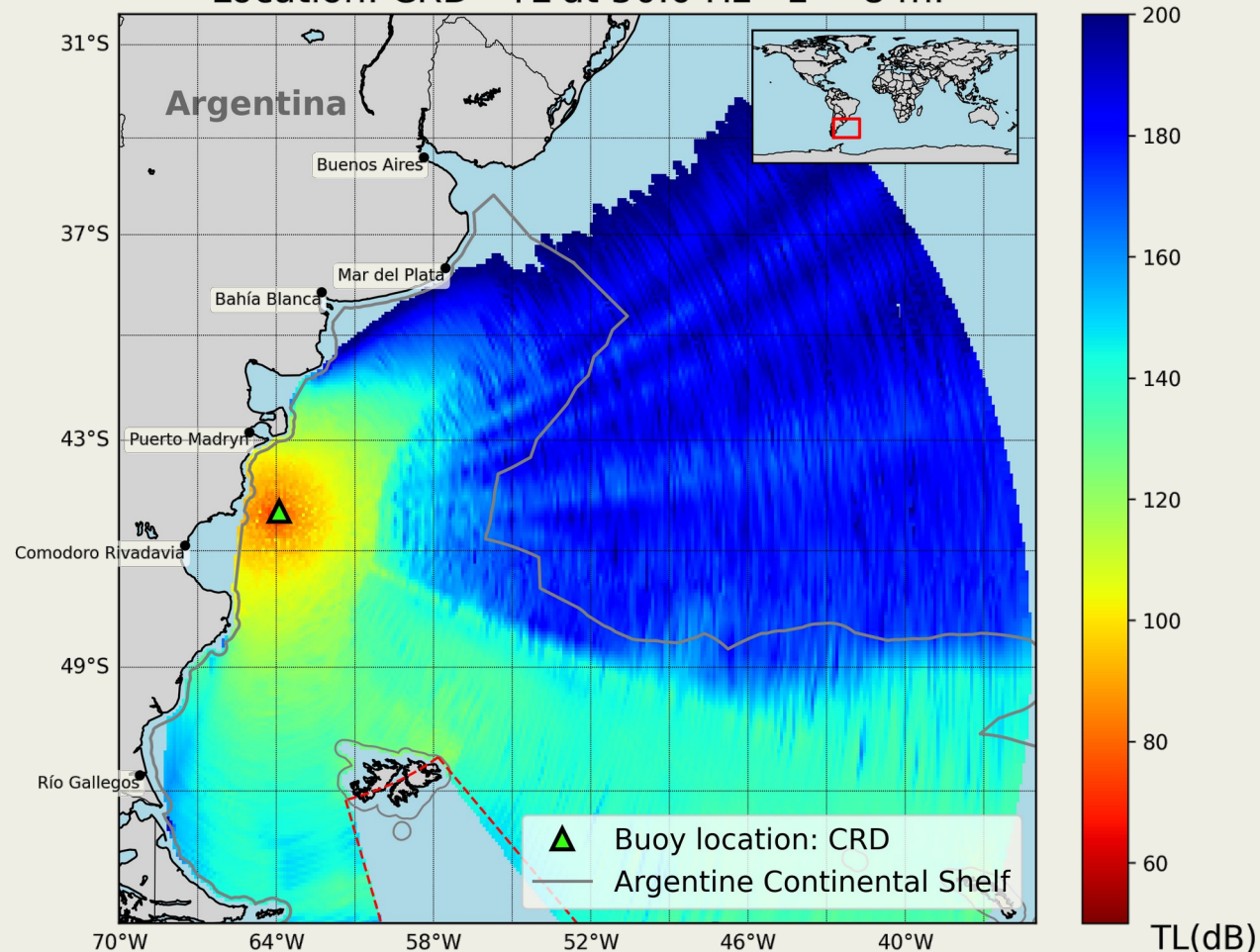
Location: CRD - TL at 20.0 Hz - $z = 8$ m.



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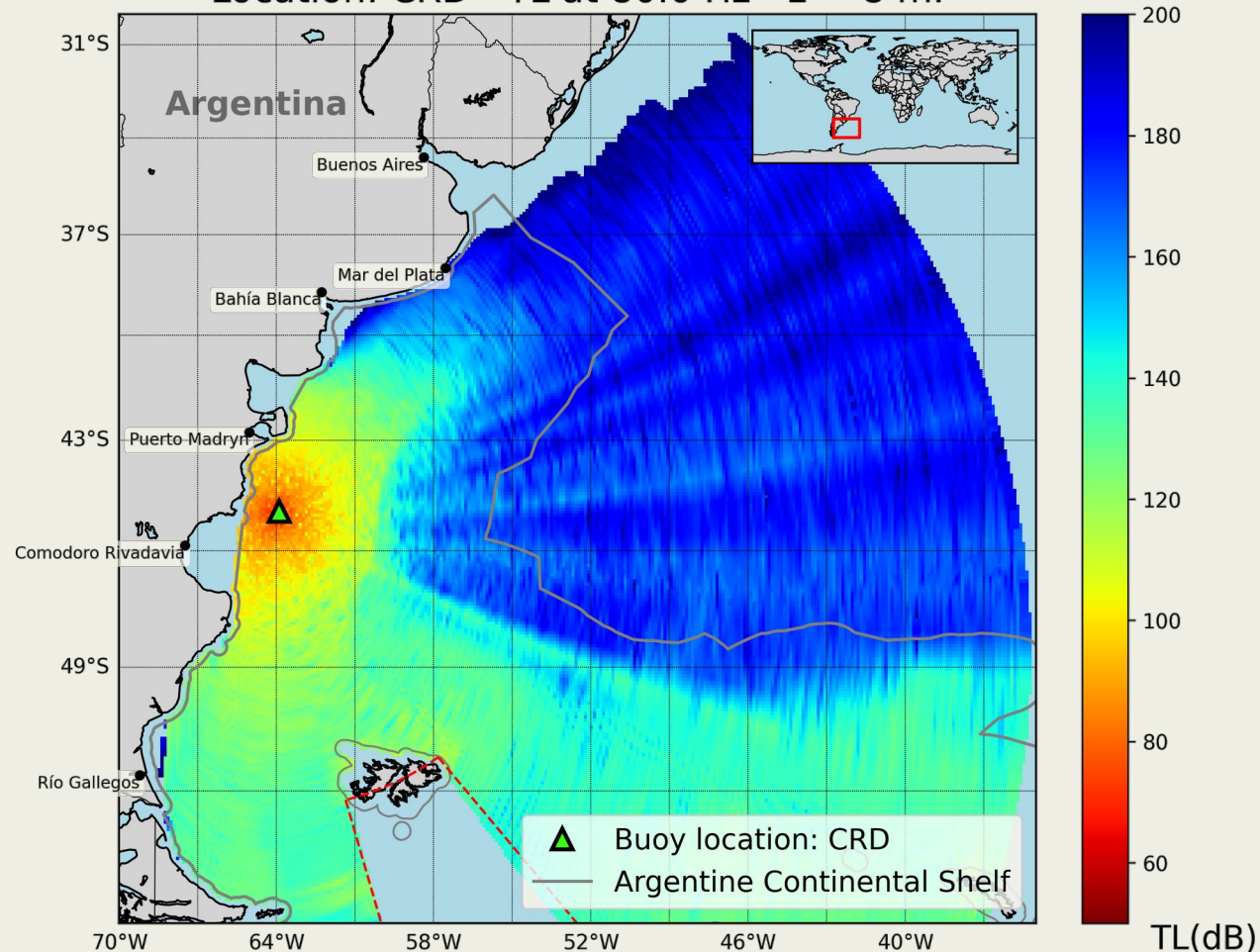
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Buoy Platform design for hydrophones deployment in selected stations

Main objectives:

- Characterization of ambient noise or underwater sound sources on the Argentinian continental shelf.
- Contributing to the understanding of shallow-water acoustic propagation.
- Potentially contribute to the understanding of long-range propagation using additional data from IMS-HA network.

This custom-designed buoy is a prototype platform developed for the deployment of hydrophone array at near-shore semi-permanent stations.



Buoy Platform design for hydrophones deployment in selected stations

Features:

- A Glass-Reinforced Plastic hull for corrosion resistance and lightweight performance.
- 3 mooring lines for stable position.
- Stand-alone operation with solar power system.
- A sealed electronics compartment and a central mast for sensors, antennas, satellite transmission and GPS.
- Harsh marine conditions: strong currents and waves up to 4 meters.



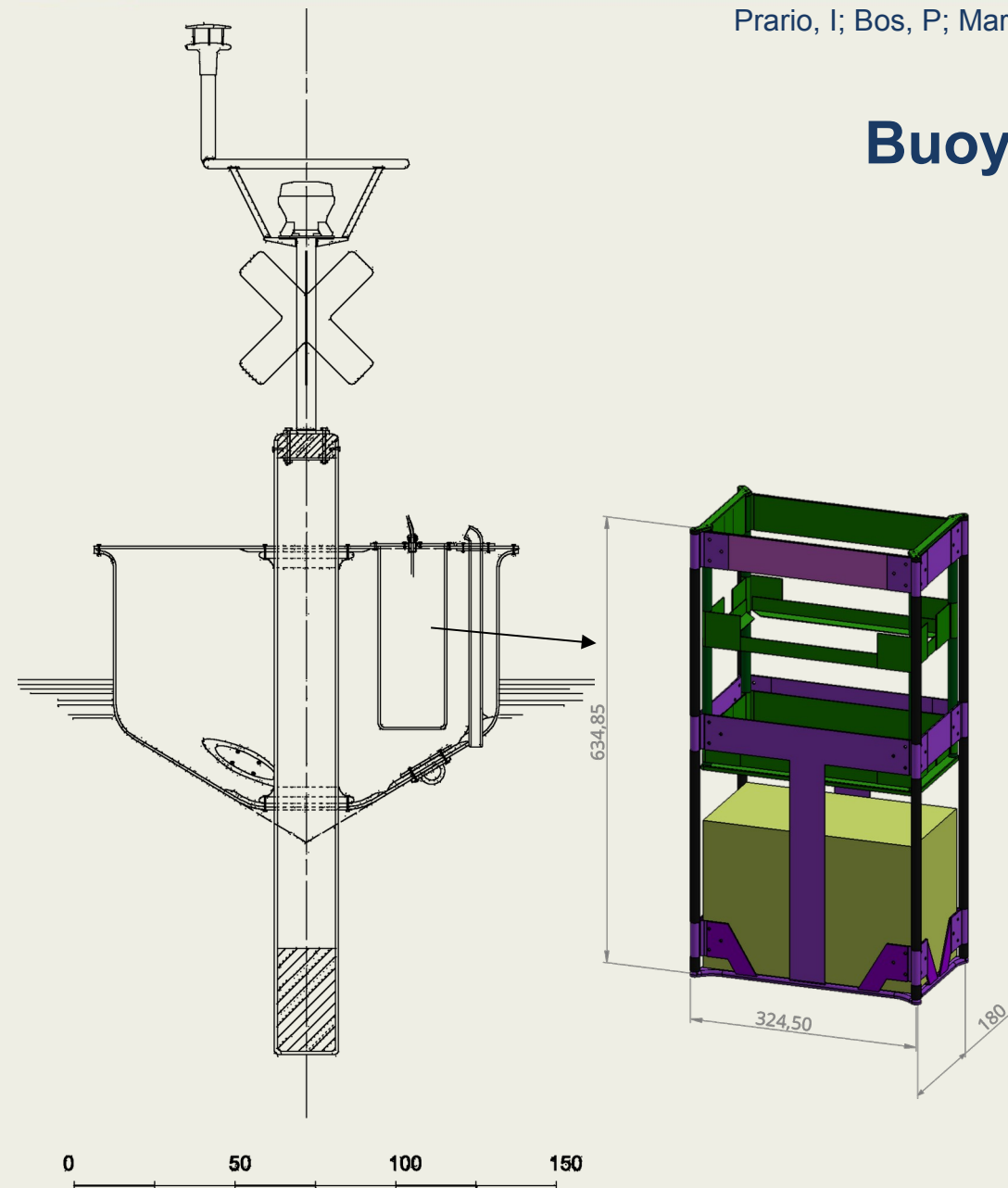
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Buoy Electronics Rack

Removable, lift-out aluminum rack houses:

- Fanless embedded computer
- 24-bit audio interface for broadband hydrophone digitization
- Low-noise hydrophone preamplifiers
- AIS-AtoN (Automatic Identification System)
- Satellite modem for telemetry and command
- Deep-cycle battery with solar charge controller



Layered Architecture of the Embedded System

Orchestrator Layer

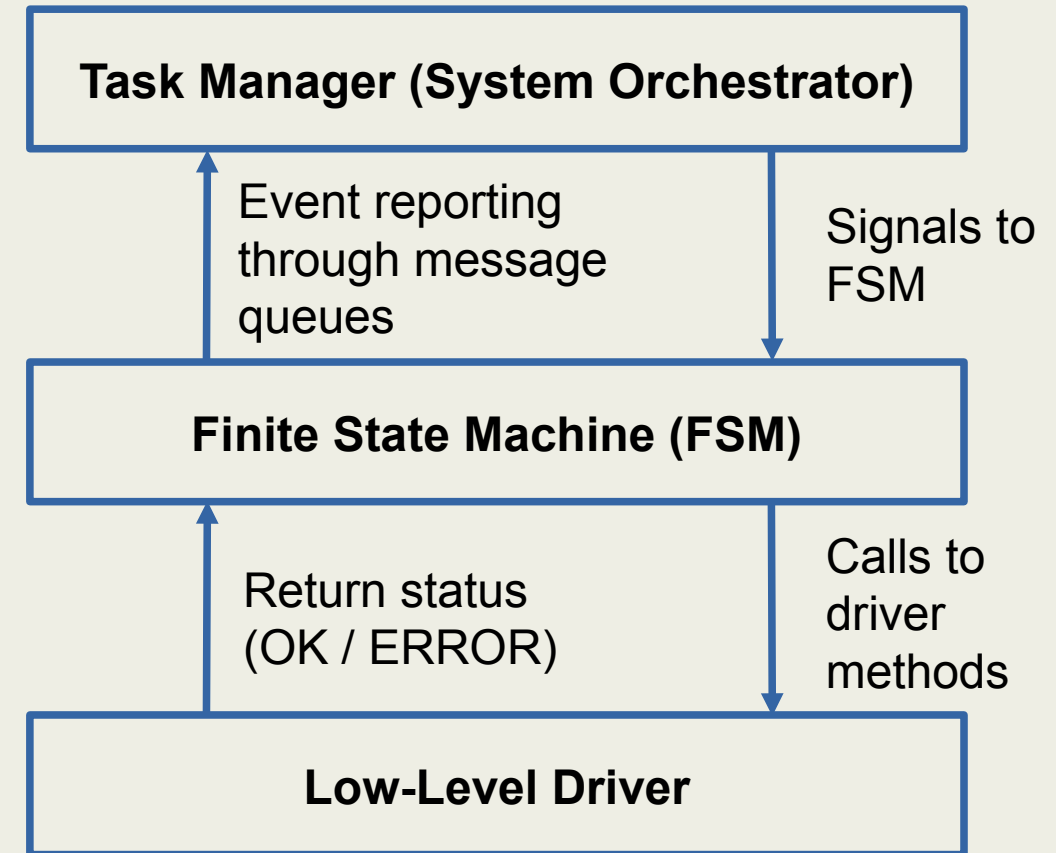
- Coordinates overall system flow.
- Decides which FSM to activate and when.
- Handles high-level internal and external events.

State Machine Layer (FSM)

- Manages module behaviour based on current state.
- React to events, executes actions, changes states.

Low-Level Driver Layer

- Interfaces directly with hardware.
- Performs device I/O and peripheral configuration.



Layered Architecture of the Embedded System

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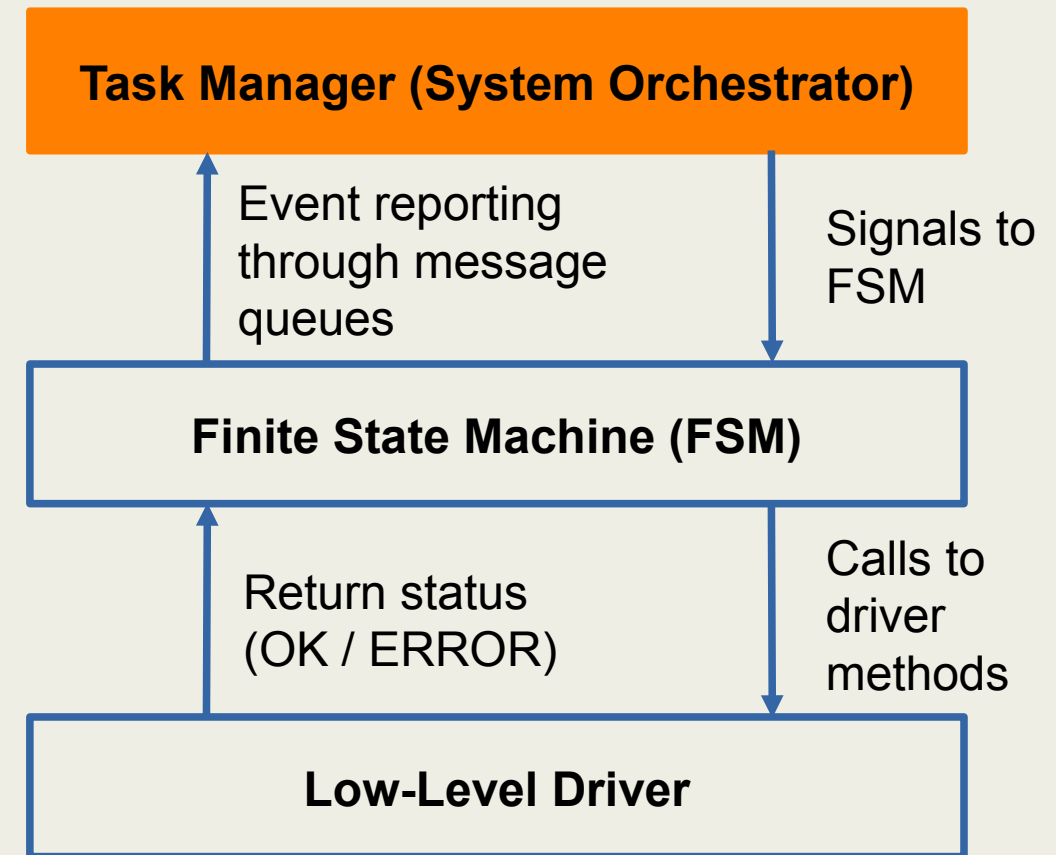
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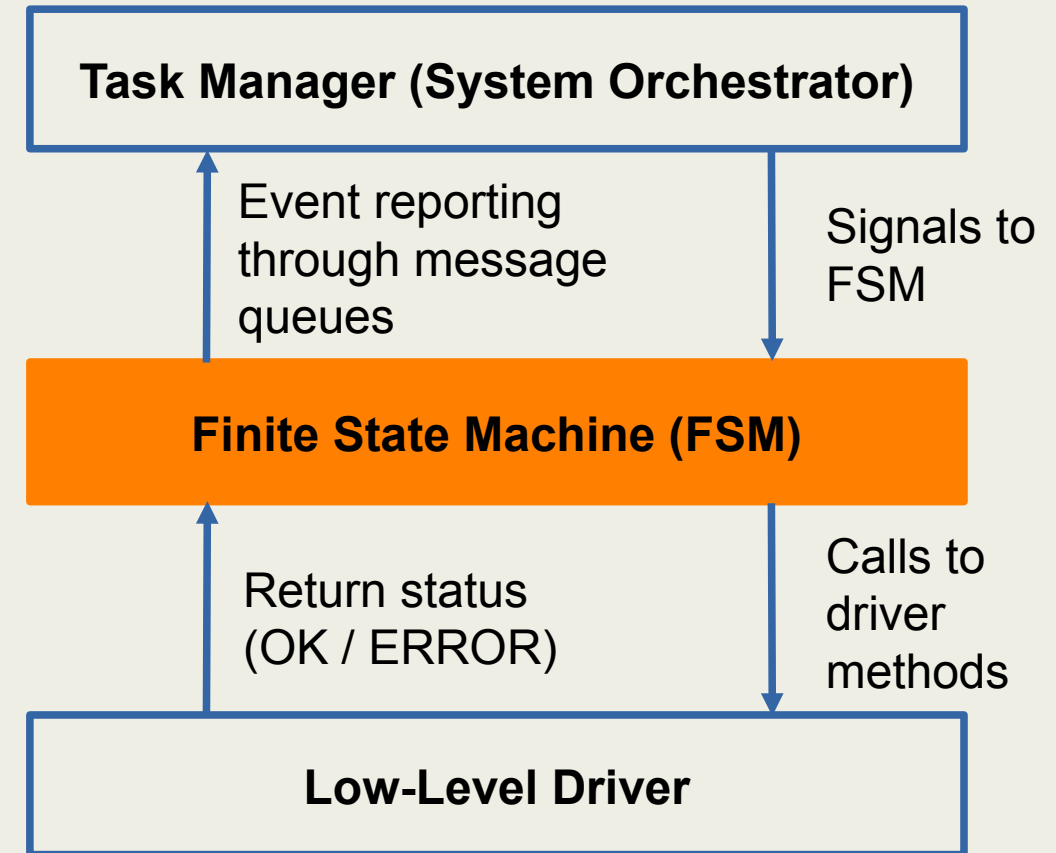
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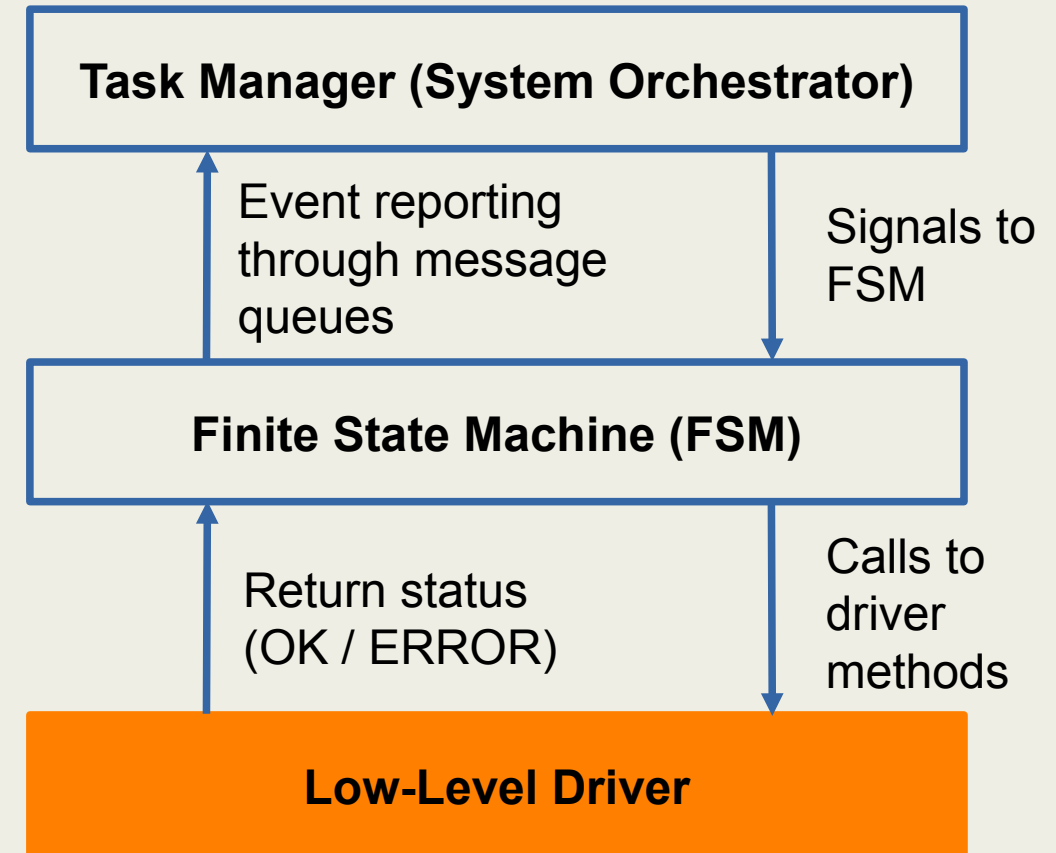
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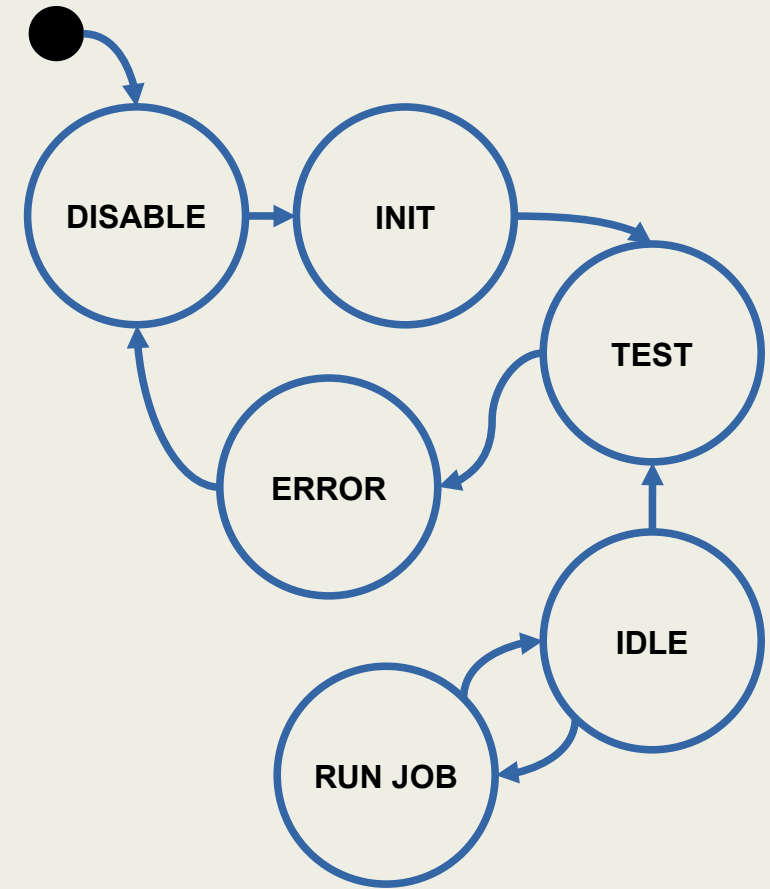
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State Machine Design Pattern

This generic FSM pattern is used across multiple modules to manage their internal logic and execution flow.

- **DISABLE**: The module is inactive and resources are released.
- **INIT**: Initializes hardware and internal variables.
- **TEST**: Performs self-checks and verifies correct functionality.
- **ERROR**: Captures faults and attempts recovery or shutdown.
- **IDLE**: Waits for commands or events.
- **RUN JOB**: Executes the main task or acquisition.



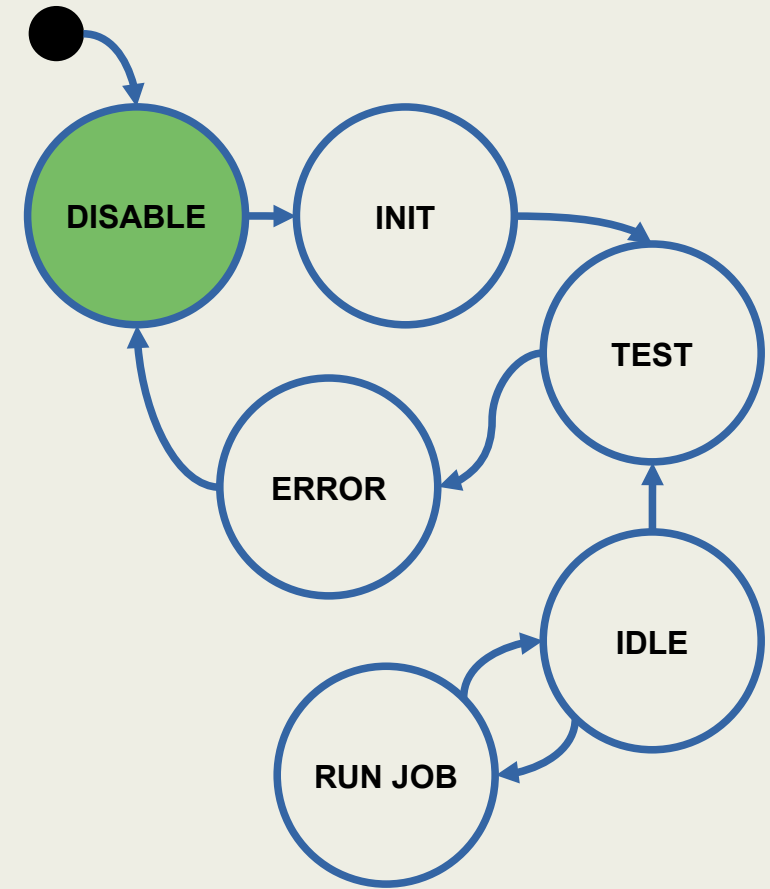
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Each FSM has a scheduler, message_handler and update functions

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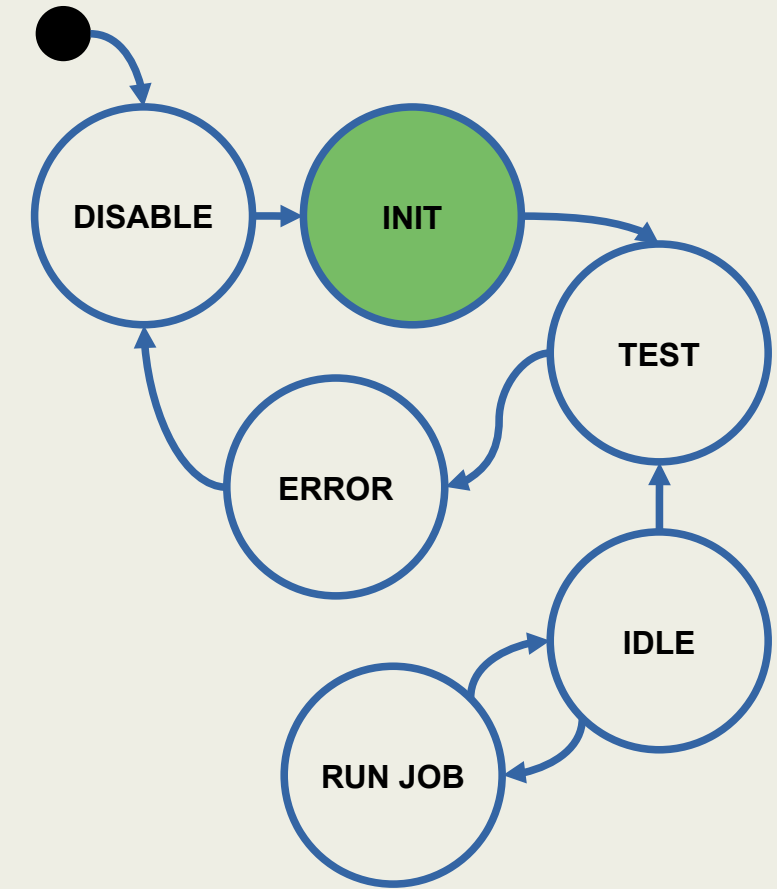
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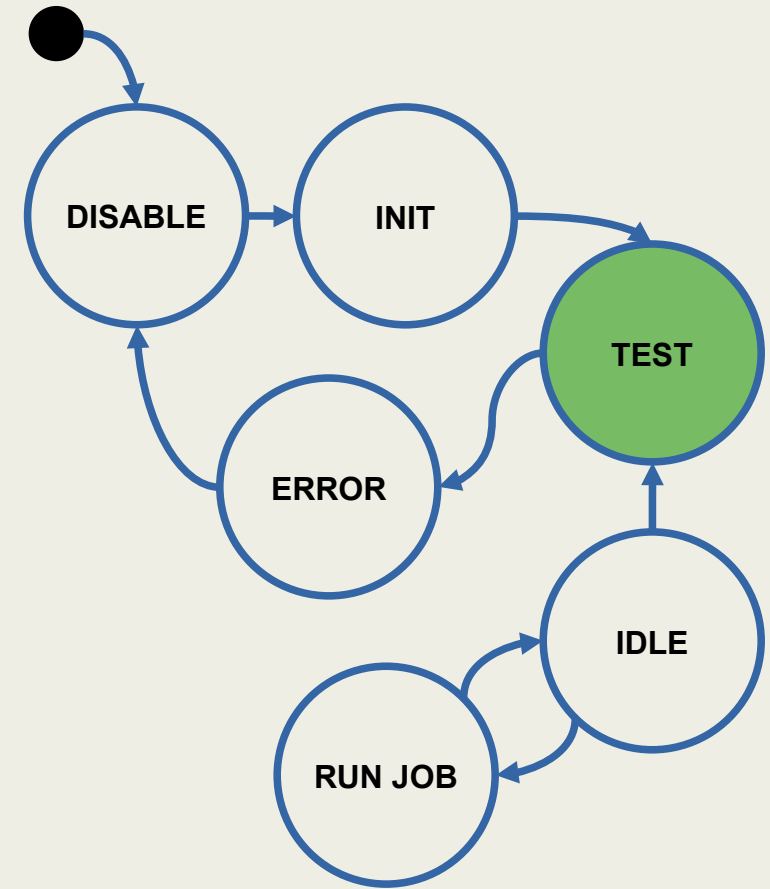
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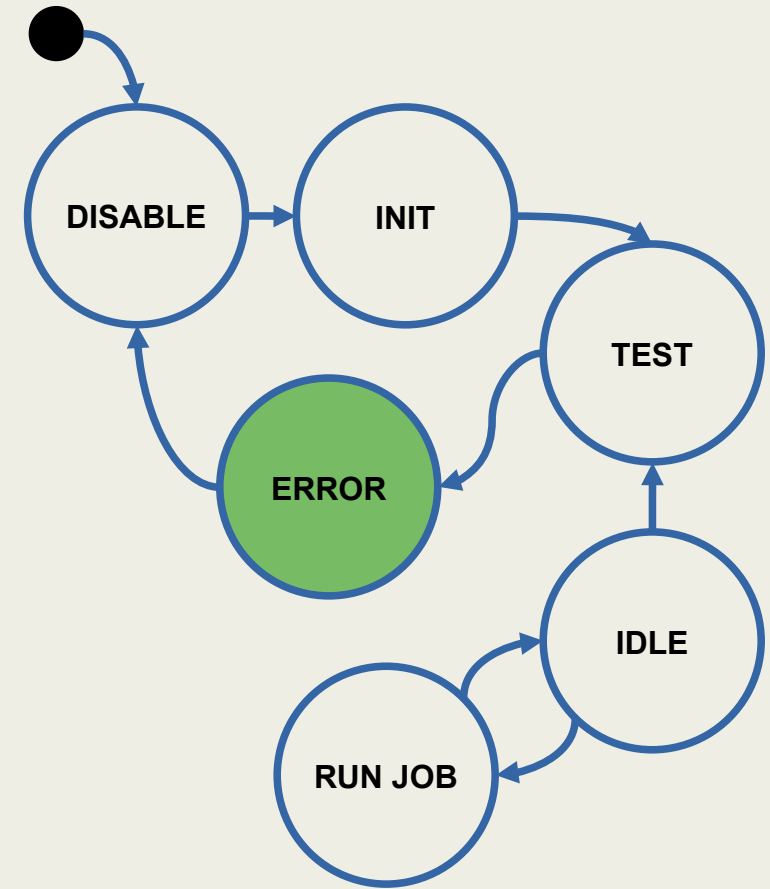
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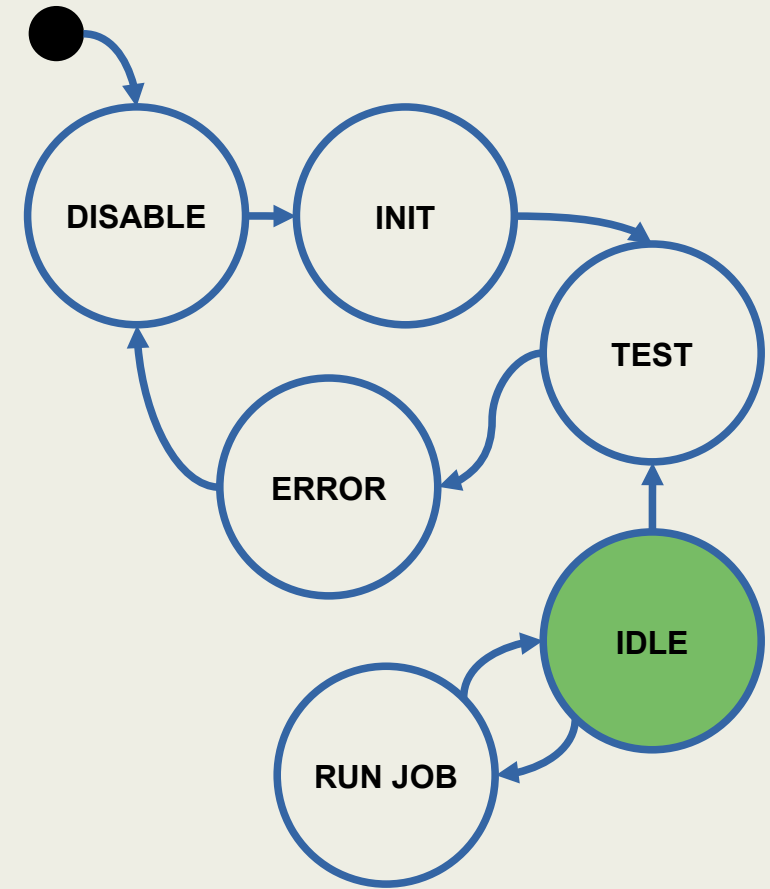
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This generic FSM pattern is used across multiple modules to manage their internal logic and execution flow.

- **DISABLE**: The module is inactive and resources are released.
- **INIT**: Initializes hardware and internal variables.
- **TEST**: Performs self-checks and verifies correct functionality.
- **ERROR**: Captures faults and attempts recovery or shutdown.
- **IDLE**: Waits for commands or events.
- **RUN JOB**: Executes the main task or acquisition.



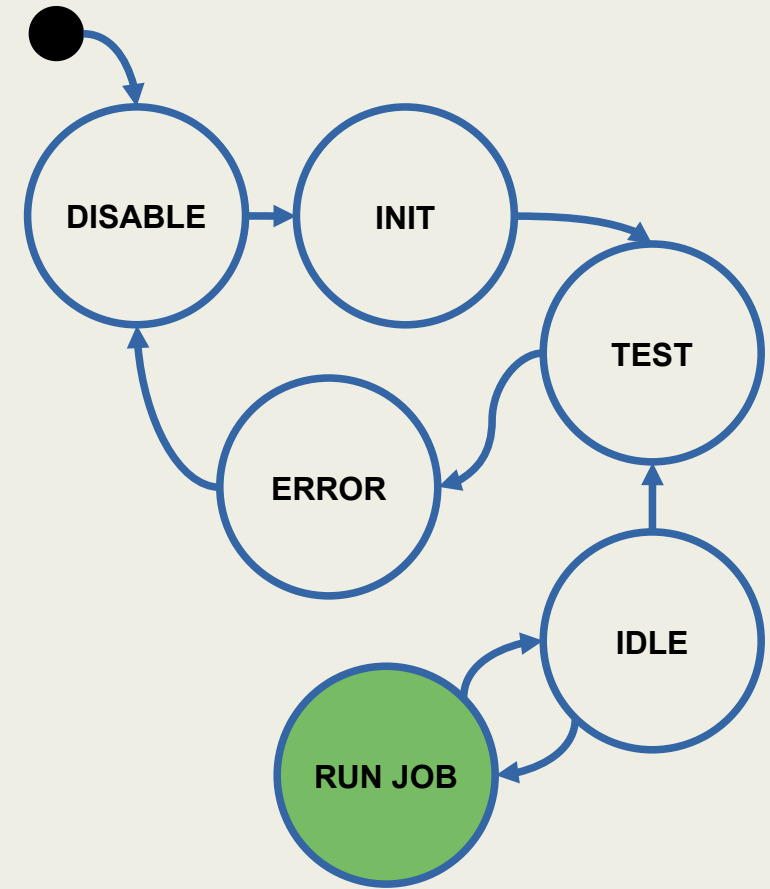
Transitions are triggered by internal events or commands.

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State Machine Design Pattern

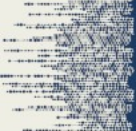
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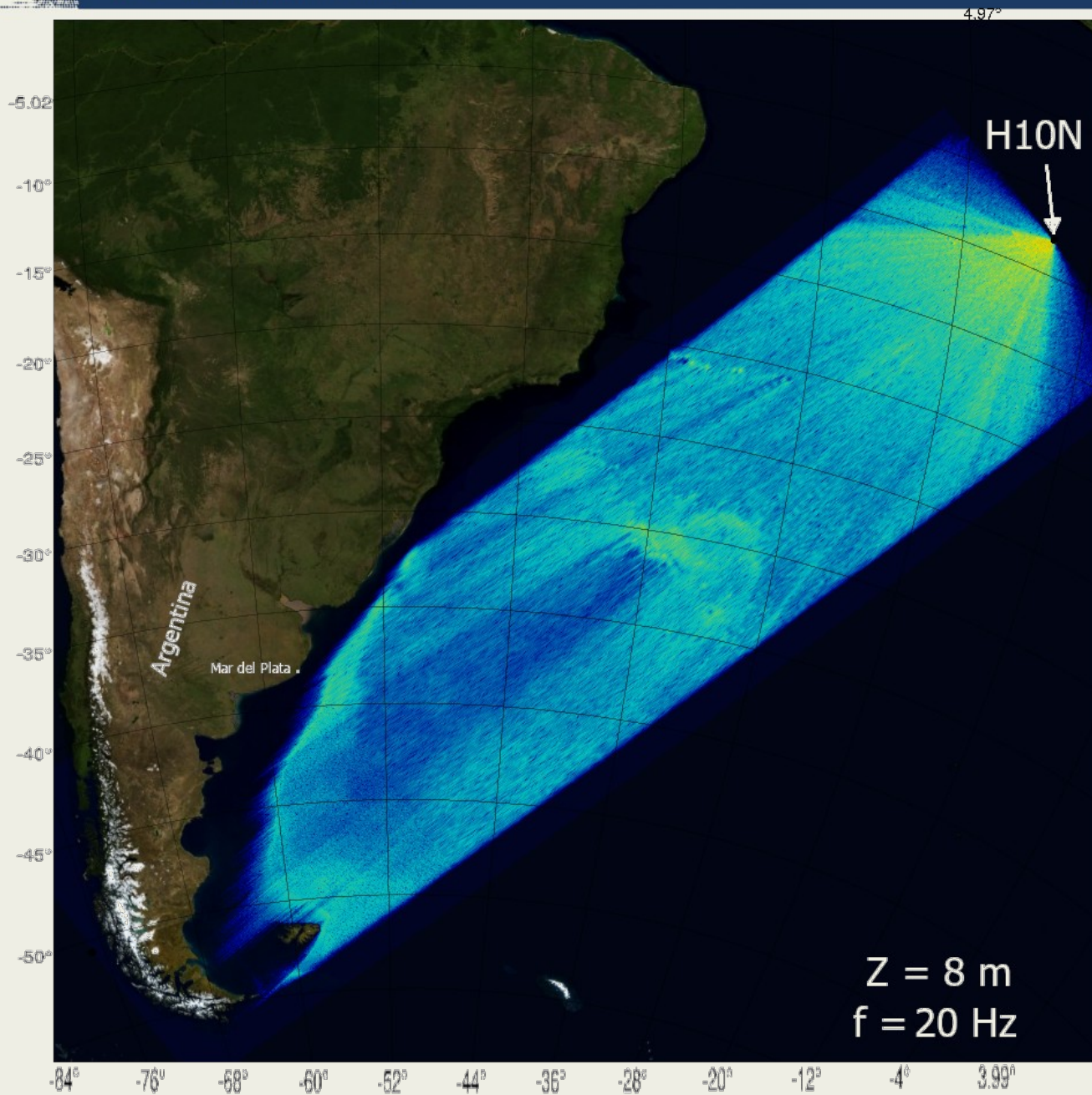
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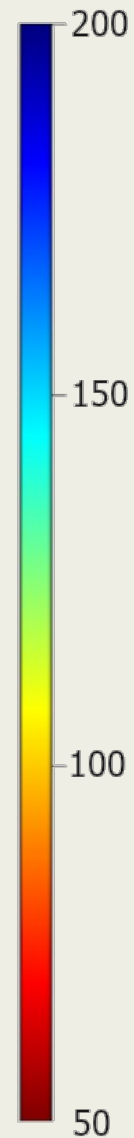


Conclusions

- ▶ A **new deployment** for underwater noise measurement was developed for **shallow waters** on the Argentinian continental shelf: application to **impulsive sound monitoring**
- ▶ Potential relevance for **basin-scale propagation analysis** using impulsive signals detected by the **IMS hydroacoustic network**.
- ▶ The **low-cost prototype buoy** for coastal **acoustic monitoring** is in its final integration stage, designed for robustness with a static mooring system.
- ▶ Proposed deployment sites (MDQ and CRD) were selected to **enhance detection** in previously **uncovered regions**.
- ▶ This platform paves the way for a **scalable coastal monitoring system** supporting baseline studies of shallow water **soundscapes** and long-range propagation analysis.
- ▶ **Coverage analysis improvement** through the **3D Split-Step Fourier Method** is currently under development; further details are available in **e-poster P1.3-766**.



TL (dB)



Thank you!